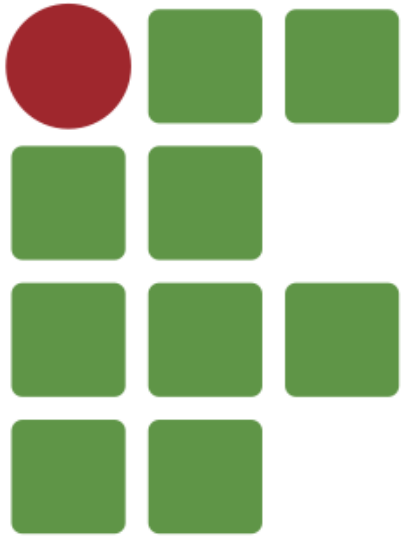


INSTITUTO FEDERAL
Brasília

ARQUITETURA DE COMPUTADORES

Prof. MSc. Thiago Neves
IFB Campus Taguatinga
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AULA 08

Assembler MIPS – Parte 2

MIPS: Vetores

Em C: vetor é um agrupamento unidimensional de elementos do mesmo tipo identificado por um índice [0, N-1].

C

char v[100];

int v[100];

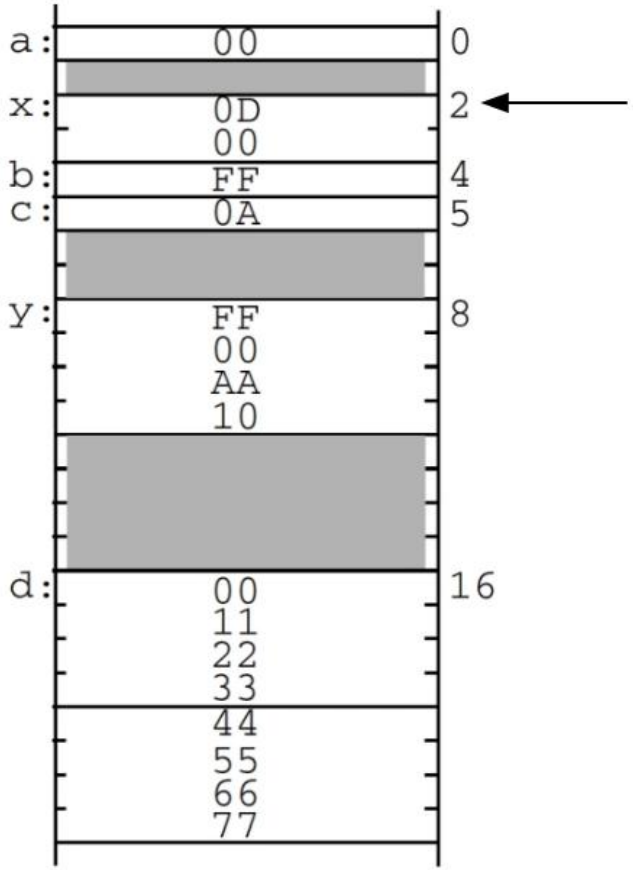
Assembler MIPS

v: .space 100

v: .space 100*4

MIPS: Alinhamento das variáveis

```
1 a:.byte 0
2 x:.half 13
3 b:.byte -1
4 c:.byte 10
5 y:.word 0x10AA00FF
6 d:.dword 0x7766554433221100
```



MIPS: Alinhamento de vetores

Align (assembler directive)

.align n: coloca o próximo dado em um endereço que é múltiplo de 2^n.

```
1  char c;
2  int v[100];
```

```
1  c:  .byte 0
2      .align 2
3  v:  .space 100*4
```

MIPS: Vetores

```
1  char c;
2  int v[100];
```

```
1  c:  .byte 0
2      .align 2
3  v:  .space 100*4
```

MIPS: Vetores

In C:

```
1  ...
2  int vec[100];
3  ...
4  int vec2[5] = {0, 1, 2, 3, 4};
```

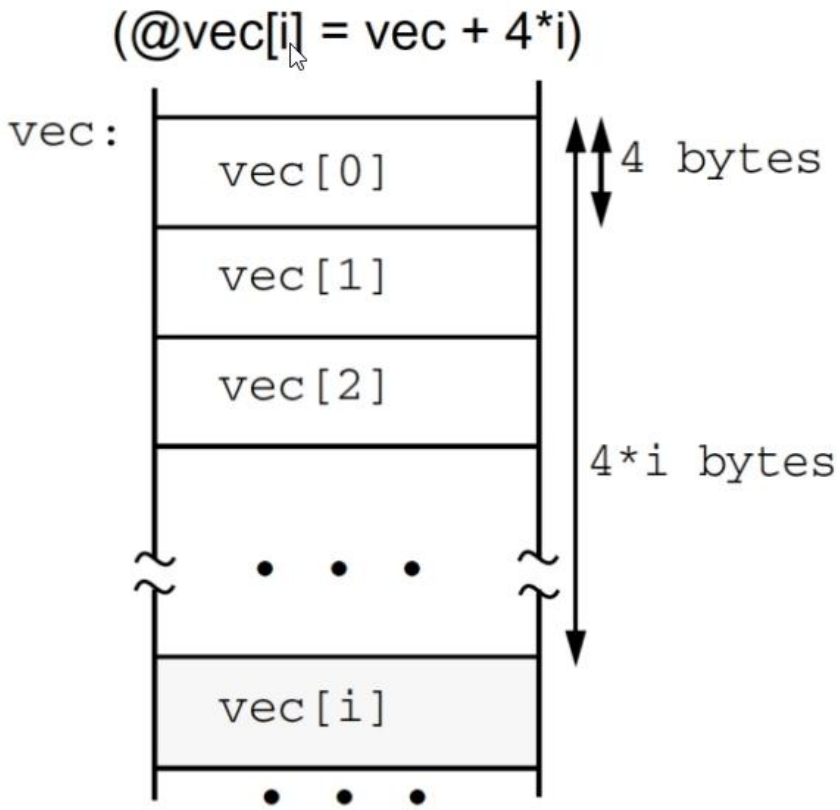
In MIPS:

```
1        .data
2  ...
3        .align 2
4  vec:  .space 400
5  ...
6  vec2:.word 0, 1, 2, 3, 4
```

Places vec in memory address multiple of 2^2

MIPS: Acesso randômico ao elemento

$$@vec[i] = @vec[0] + i \cdot T$$



Shift Left Logical (ssl)

	$t1	shamt	$t0	
consider:	0001	1	0010	*2
	0001	2	0100	*4
	0001	3	1000	*8

shift left logical is multiplication by 2^{shamt}

MIPS: Acesso randômico ao elemento

```
1  int vec[100];
2  main()
3  {
4     int x, i; //    $t1 , $t2  respectively
5     ...
6     x = vec[i];
7  }
```

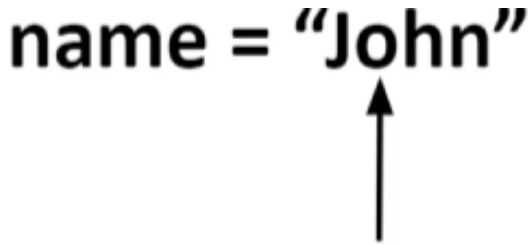
```
1        #&vec[i] = &vec[0] + i*4
2        la   $t0, vec          #&vec[0]
3        sll  $t3, $t2, 2       #$t3 = i<<2 = i * 4
4        addu $t0, $t0, $t3     #$t0 = &vec[0] + i*4
5
6        lw   $t1, 0($t0)       #x = vec[i]
```

Strings em C: declaração

char name[80] = “Instituto Federal de Brasilia”; // Adiciona o ‘\0’

char name2[80] = {‘I’, ‘F’, ‘B’, ‘\0’};

Strings em C: acesso à elementos




```
char name[80] ;
main()  {
int num = 0;
…
while (name[num] != ‘\0’) num += 1 ;
…
}
```

MIPS: escrevendo um while

```
char name[80] ;
main() {
int num = 0;
...
while (name[num] != '\0') num += 1 ;
...
}
```

A[i] → @A[0] + i * T

name[num] → @name[0] + num * 1

```
1  main : . . .
2     move   $t0 , $zero      # num = 0
3     la     $t1 , nom
4  while :
5     addu   $t3 , $t1 , $t0  # $t3 = @name[0] + num*1
6     lb     $t2 , 0($t3)     # $t2 = name[num]
7     beq    $t2 , $zero ,    endwhile
8     addiu  $t0 , $t0 , 1    # num += 1
9     b  while
10 endwhile:
11 . . .
```

MARS - syscall

Em Assembly MIPS, os serviços de syscall permitem imprimir valores, ler dados digitados pelo usuário, manipular arquivos, alocar memória e encerrar o programa.

MARS: Como utilizar um serviço

A sequência padrão é carregar o número do serviço em $v0, preparar os argumentos nos registradores indicados e executar syscall.

Exemplo:

```
li   $v0, 1          # Serviço 1: imprimir inteiro
li   $a0, 42         # Valor que será impresso
syscall
```

MARS: Lista dos serviços básicos (1 a 17)

Código em $v0	Serviço	Argumentos / resultado
1	Imprimir inteiro	$a0 = inteiro
2	Imprimir float	$f12 = número real
3	Imprimir double	$f12 = número de ponto flutuante de dupla precisão
4	Imprimir string	$a0 = endereço da string
5	Ler inteiro	Retorno em $v0
6	Ler float	Retorno em $f0
7	Ler double	Retorno em $f0 e $f1
8	Ler string	$a0 = endereço do buffer; $a1 = tamanho máximo
9	Alocar memória dinâmica	$a0 = bytes solicitados; endereço retornado em $v0
10	Encerrar programa	Sem argumentos
11	Imprimir caractere	$a0 = código ASCII
12	Ler caractere	Caractere retornado em $v0
13	Abrir arquivo	$a0 = nome; $a1 = flags; $a2 = modo
14	Ler arquivo	$a0 = descritor; $a1 = buffer; $a2 = quantidade máxima
15	Escrever arquivo	$a0 = descritor; $a1 = buffer; $a2 = quantidade
16	Fechar arquivo	$a0 = descritor
17	Encerrar com código de saída	$a0 = código de saída

MARS: Serviços adicionais

Código	Serviço	Finalidade
30	Time	Obter o horário do sistema em milissegundos
31	MIDI out	Reproduzir um som sem esperar terminar
32	Sleep	Pausar a execução por um intervalo
33	MIDI out síncrono	Reproduzir um som e esperar terminar
34	Imprimir inteiro hexadecimal	Exibir o valor em hexadecimal
35	Imprimir inteiro binário	Exibir os 32 bits do valor
36	Imprimir inteiro sem sinal	Exibir inteiro como unsigned
37–39	Não utilizados	—
40	Definir semente aleatória	Inicializar gerador de números aleatórios
41	Gerar inteiro aleatório	Gerar inteiro pseudoaleatório
42	Gerar inteiro aleatório em faixa	Gerar inteiro de 0 até limite exclusivo
43	Gerar float aleatório	Gerar número entre 0 e 1
44	Gerar double aleatório	Gerar número entre 0 e 1
45–49	Não utilizados	—
50	Caixa de confirmação	Exibir opções Sim, Não e Cancelar
51	Caixa de entrada de inteiro	Solicitar um inteiro em uma janela
52	Caixa de entrada de float	Solicitar um número real
53	Caixa de entrada de double	Solicitar número de dupla precisão
54	Caixa de entrada de string	Solicitar texto em uma janela
55	Caixa de mensagem	Exibir mensagem em uma janela
56	Mensagem com inteiro	Exibir mensagem acompanhada de inteiro
57	Mensagem com float	Exibir mensagem acompanhada de float
58	Mensagem com double	Exibir mensagem acompanhada de double
59	Mensagem com string	Exibir mensagem acompanhada de texto

Tabela ASCII

Dec	Hex	Oct	Chr	Dec	Hex	Oct	HTML	Chr	Dec	Hex	Oct	HTML	Chr	Dec	Hex	Oct	HTML	Chr
0	0	000	NULL	32	20	040		Space	64	40	100	@	@	96	60	140	`	`
1	1	001	Start of Header	33	21	041	!	!	65	41	101	A	A	97	61	141	a	a
2	2	002	Start of Text	34	22	042	"	"	66	42	102	B	B	98	62	142	b	b
3	3	003	End of Text	35	23	043	#	#	67	43	103	C	C	99	63	143	c	c
4	4	004	End of Transmission	36	24	044	$	$	68	44	104	D	D	100	64	144	d	d
5	5	005	Enquiry	37	25	045	%	%	69	45	105	E	E	101	65	145	e	e
6	6	006	Acknowledgment	38	26	046	&	&	70	46	106	F	F	102	66	146	f	f
7	7	007	Bell	39	27	047	'	'	71	47	107	G	G	103	67	147	g	g
8	8	010	Backspace	40	28	050	(	(	72	48	110	H	H	104	68	150	h	h
9	9	011	Horizontal Tab	41	29	051	)	)	73	49	111	I	I	105	69	151	i	i
10	A	012	Line feed	42	2A	052	*	*	74	4A	112	J	J	106	6A	152	j	j
11	B	013	Vertical Tab	43	2B	053	+	+	75	4B	113	K	K	107	6B	153	k	k
12	C	014	Form feed	44	2C	054	,	,	76	4C	114	L	L	108	6C	154	l	l
13	D	015	Carriage return	45	2D	055	-	-	77	4D	115	M	M	109	6D	155	m	m
14	E	016	Shift Out	46	2E	056	.	.	78	4E	116	N	N	110	6E	156	n	n
15	F	017	Shift In	47	2F	057	/	/	79	4F	117	O	O	111	6F	157	o	o
16	10	020	Data Link Escape	48	30	060	0	0	80	50	120	P	P	112	70	160	p	p
17	11	021	Device Control 1	49	31	061	1	1	81	51	121	Q	Q	113	71	161	q	q
18	12	022	Device Control 2	50	32	062	2	2	82	52	122	R	R	114	72	162	r	r
19	13	023	Device Control 3	51	33	063	3	3	83	53	123	S	S	115	73	163	s	s
20	14	024	Device Control 4	52	34	064	4	4	84	54	124	T	T	116	74	164	t	t
21	15	025	Negative Ack.	53	35	065	5	5	85	55	125	U	U	117	75	165	u	u
22	16	026	Synchronous idle	54	36	066	6	6	86	56	126	V	V	118	76	166	v	v
23	17	027	End of Trans. Block	55	37	067	7	7	87	57	127	W	W	119	77	167	w	w
24	18	030	Cancel	56	38	070	8	8	88	58	130	X	X	120	78	170	x	x
25	19	031	End of Medium	57	39	071	9	9	89	59	131	Y	Y	121	79	171	y	y
26	1A	032	Substitute	58	3A	072	:	:	90	5A	132	Z	Z	122	7A	172	z	z
27	1B	033	Escape	59	3B	073	;	;	91	5B	133	[	[	123	7B	173	{	{
28	1C	034	File Separator	60	3C	074	<	<	92	5C	134	\	\	124	7C	174			|
29	1D	035	Group Separator	61	3D	075	=	=	93	5D	135	]	]	125	7D	175	}	}
30	1E	036	Record Separator	62	3E	076	>	>	94	5E	136	^	^	126	7E	176	~	~
31	1F	037	Unit Separator	63	3F	077	?	?	95	5F	137	_	_	127	7F	177	&#127;	Del

asciicharstable.com

Exemplo 1 - MARS

O exemplo a seguir mostra a execução de código MIPS no simulador MARS.

Exemplo 1

```
1  vetor: .byte 10, 20, 30, 0
2
3  .text
4  .globl main
5
6  main:
7          move $t0, $zero
8          la $t1, vetor
9  while:
10         addu $t3, $t1, $t0
11         lb $t2, 0($t3)
12
13         # Serviço 1 - Imprimir o inteiro que estiver no registrador $a0.
14         li $v0, 1
15         move $a0, $t2
16         syscall
17         # Servico 11 - Imprime o caractere que estiver no registrador $a0 (ANCII)
18         li $v0, 11
19         addiu $a0, $zero, 32
20         syscall
21
22         # While $t2 != 0
23         beq $t2, $zero, fim
24         addiu $t0, $t0, 1
25         b while
26
27 fim:    # Serviço 10 - Encerrar o programa.
28         li $v0, 10
29         syscall
30
```

Passo 1

Text Segment

Bkpt	Address	Code	Basic		Source
	0x00400000	0x00004021	addu $8,$0,$0	8:	move $t0, $zero
	0x00400004	0x3c011001	lui $1,0x00001001	9:	la $t1, vetor
	0x00400008	0x34290000	ori $9,$1,0x0000...		
	0x0040000c	0x01285821	addu $11,$9,$8	11:	addu $t3, $t1, $t0
	0x00400010	0x816a0000	lb $10,0x0000000...	12:	lb $t2, 0($t3)
	0x00400014	0x24020001	addiu $2,$0,0x00...	15:	li $v0, 1
	0x00400018	0x000a2021	addu $4,$0,$10	16:	move $a0, $t2
	0x0040001c	0x0000000c	syscall	17:	syscall
	0x00400020	0x2402000b	addiu $2,$0,0x00...	19:	li $v0, 11
	0x00400024	0x24040020	addiu $4,$0,0x00...	20:	addiu $a0, $zero, 32
	0x00400028	0x0000000c	syscall	21:	syscall
	0x0040002c	0x11400002	beq $10,$0,0x000...	24:	beq $t2, $zero, fim
	0x00400030	0x25080001	addiu $8,$8,0x00...	25:	addiu $t0, $t0, 1
	0x00400034	0x0401fff5	bgez $0,0xfffffff5	26:	b while
	0x00400038	0x2402000a	addiu $2,$0,0x00...	29:	li $v0, 10
	0x0040003c	0x0000000c	syscall	30:	syscall

Name	Number	Value
$zero	0	0x00000000
$at	1	0x00000000
$v0	2	0x00000000
$v1	3	0x00000000
$a0	4	0x00000000
$a1	5	0x00000000
$a2	6	0x00000000
$a3	7	0x00000000
$t0	8	0x00000000
$t1	9	0x00000000
$t2	10	0x00000000
$t3	11	0x00000000
$t4	12	0x00000000
$t5	13	0x00000000
$t6	14	0x00000000
$t7	15	0x00000000
$s0	16	0x00000000
$s1	17	0x00000000
$s2	18	0x00000000
$s3	19	0x00000000
$s4	20	0x00000000
$s5	21	0x00000000
$s6	22	0x00000000
$s7	23	0x00000000
$t8	24	0x00000000
$t9	25	0x00000000
$k0	26	0x00000000
$k1	27	0x00000000
$gp	28	0x10008000
$sp	29	0x7fffeffc
$fp	30	0x00000000
$ra	31	0x00000000
pc		0x00400000
hi		0x00000000
lo		0x00000000

Passo 2

Text Segment

Bkpt	Address	Code	Basic		Source
	0x00400000	0x00004021	addu $8,$0,$0	8:	move $t0, $zero
	0x00400004	0x3c011001	lui $1,0x00001001	9:	la $t1, vetor
	0x00400008	0x34290000	ori $9,$1,0x0000...		
	0x0040000c	0x01285821	addu $11,$9,$8	11:	addu $t3, $t1, $t0
	0x00400010	0x816a0000	lb $10,0x0000000...	12:	lb $t2, 0($t3)
	0x00400014	0x24020001	addiu $2,$0,0x00...	15:	li $v0, 1
	0x00400018	0x000a2021	addu $4,$0,$10	16:	move $a0, $t2
	0x0040001c	0x0000000c	syscall	17:	syscall
	0x00400020	0x2402000b	addiu $2,$0,0x00...	19:	li $v0, 11
	0x00400024	0x24040020	addiu $4,$0,0x00...	20:	addiu $a0, $zero, 32
	0x00400028	0x0000000c	syscall	21:	syscall
	0x0040002c	0x11400002	beq $10,$0,0x000...	24:	beq $t2, $zero, fim
	0x00400030	0x25080001	addiu $8,$8,0x00...	25:	addiu $t0, $t0, 1
	0x00400034	0x0401fff5	bgez $0,0xfffffff5	26:	b while
	0x00400038	0x2402000a	addiu $2,$0,0x00...	29:	li $v0, 10
	0x0040003c	0x0000000c	syscall	30:	syscall

Name	Number	Value
$zero	0	0x00000000
$at	1	0x00000000
$v0	2	0x00000000
$v1	3	0x00000000
$a0	4	0x00000000
$a1	5	0x00000000
$a2	6	0x00000000
$a3	7	0x00000000
$t0	8	0x00000000
$t1	9	0x00000000
$t2	10	0x00000000
$t3	11	0x00000000
$t4	12	0x00000000
$t5	13	0x00000000
$t6	14	0x00000000
$t7	15	0x00000000
$s0	16	0x00000000
$s1	17	0x00000000
$s2	18	0x00000000
$s3	19	0x00000000
$s4	20	0x00000000
$s5	21	0x00000000
$s6	22	0x00000000
$s7	23	0x00000000
$t8	24	0x00000000
$t9	25	0x00000000
$k0	26	0x00000000
$k1	27	0x00000000
$gp	28	0x10008000
$sp	29	0x7fffeffc
$fp	30	0x00000000
$ra	31	0x00000000
pc		0x00400004
hi		0x00000000
lo		0x00000000

Passo 3

Text Segment

Bkpt	Address	Code	Basic		Source
	0x00400000	0x00004021	addu $8,$0,$0	8:	move $t0, $zero
	0x00400004	0x3c011001	lui $1,0x00001001	9:	la $t1, vetor
	0x00400008	0x34290000	ori $9,$1,0x0000...		
	0x0040000c	0x01285821	addu $11,$9,$8	11:	addu $t3, $t1, $t0
	0x00400010	0x816a0000	lb $10,0x0000000...	12:	lb $t2, 0($t3)
	0x00400014	0x24020001	addiu $2,$0,0x00...	15:	li $v0, 1
	0x00400018	0x000a2021	addu $4,$0,$10	16:	move $a0, $t2
	0x0040001c	0x0000000c	syscall	17:	syscall
	0x00400020	0x2402000b	addiu $2,$0,0x00...	19:	li $v0, 11
	0x00400024	0x24040020	addiu $4,$0,0x00...	20:	addiu $a0, $zero, 32
	0x00400028	0x0000000c	syscall	21:	syscall
	0x0040002c	0x11400002	beq $10,$0,0x000...	24:	beq $t2, $zero, fim
	0x00400030	0x25080001	addiu $8,$8,0x00...	25:	addiu $t0, $t0, 1
	0x00400034	0x0401fff5	bgez $0,0xfffffff5	26:	b while
	0x00400038	0x2402000a	addiu $2,$0,0x00...	29:	li $v0, 10
	0x0040003c	0x0000000c	syscall	30:	syscall

Name	Number	Value
$zero	0	0x00000000
$at	1	0x10010000
$v0	2	0x00000000
$v1	3	0x00000000
$a0	4	0x00000000
$a1	5	0x00000000
$a2	6	0x00000000
$a3	7	0x00000000
$t0	8	0x00000000
$t1	9	0x00000000
$t2	10	0x00000000
$t3	11	0x00000000
$t4	12	0x00000000
$t5	13	0x00000000
$t6	14	0x00000000
$t7	15	0x00000000
$s0	16	0x00000000
$s1	17	0x00000000
$s2	18	0x00000000
$s3	19	0x00000000
$s4	20	0x00000000
$s5	21	0x00000000
$s6	22	0x00000000
$s7	23	0x00000000
$t8	24	0x00000000
$t9	25	0x00000000
$k0	26	0x00000000
$k1	27	0x00000000
$gp	28	0x10008000
$sp	29	0x7fffeffc
$fp	30	0x00000000
$ra	31	0x00000000
pc		0x00400008
hi		0x00000000
lo		0x00000000

Passo 4

Text Segment

Bkpt	Address	Code	Basic		Source
	0x00400000	0x00004021	addu $8,$0,$0	8:	move $t0, $zero
	0x00400004	0x3c011001	lui $1,0x00001001	9:	la $t1, vetor
	0x00400008	0x34290000	ori $9,$1,0x0000...		
	0x0040000c	0x01285821	addu $11,$9,$8	11:	addu $t3, $t1, $t0
	0x00400010	0x816a0000	lb $10,0x0000000...	12:	lb $t2, 0($t3)
	0x00400014	0x24020001	addiu $2,$0,0x00...	15:	li $v0, 1
	0x00400018	0x000a2021	addu $4,$0,$10	16:	move $a0, $t2
	0x0040001c	0x0000000c	syscall	17:	syscall
	0x00400020	0x2402000b	addiu $2,$0,0x00...	19:	li $v0, 11
	0x00400024	0x24040020	addiu $4,$0,0x00...	20:	addiu $a0, $zero, 32
	0x00400028	0x0000000c	syscall	21:	syscall
	0x0040002c	0x11400002	beq $10,$0,0x000...	24:	beq $t2, $zero, fim
	0x00400030	0x25080001	addiu $8,$8,0x00...	25:	addiu $t0, $t0, 1
	0x00400034	0x0401fff5	bgez $0,0xfffffff5	26:	b while
	0x00400038	0x2402000a	addiu $2,$0,0x00...	29:	li $v0, 10
	0x0040003c	0x0000000c	syscall	30:	syscall

Name	Number	Value
$zero	0	0x00000000
$at	1	0x10010000
$v0	2	0x00000000
$v1	3	0x00000000
$a0	4	0x00000000
$a1	5	0x00000000
$a2	6	0x00000000
$a3	7	0x00000000
$t0	8	0x00000000
$t1	9	0x10010000
$t2	10	0x00000000
$t3	11	0x00000000
$t4	12	0x00000000
$t5	13	0x00000000
$t6	14	0x00000000
$t7	15	0x00000000
$s0	16	0x00000000
$s1	17	0x00000000
$s2	18	0x00000000
$s3	19	0x00000000
$s4	20	0x00000000
$s5	21	0x00000000
$s6	22	0x00000000
$s7	23	0x00000000
$t8	24	0x00000000
$t9	25	0x00000000
$k0	26	0x00000000
$k1	27	0x00000000
$gp	28	0x10008000
$sp	29	0x7fffeffc
$fp	30	0x00000000
$ra	31	0x00000000
pc		0x0040000c
hi		0x00000000
lo		0x00000000

Passo 5

Text Segment

Bkpt	Address	Code	Basic		Source
	0x00400000	0x00004021	addu $8,$0,$0	8:	move $t0, $zero
	0x00400004	0x3c011001	lui $1,0x00001001	9:	la $t1, vetor
	0x00400008	0x34290000	ori $9,$1,0x0000...		
	0x0040000c	0x01285821	addu $11,$9,$8	11:	addu $t3, $t1, $t0
	0x00400010	0x816a0000	lb $10,0x0000000...	12:	lb $t2, 0($t3)
	0x00400014	0x24020001	addiu $2,$0,0x00...	15:	li $v0, 1
	0x00400018	0x000a2021	addu $4,$0,$10	16:	move $a0, $t2
	0x0040001c	0x0000000c	syscall	17:	syscall
	0x00400020	0x2402000b	addiu $2,$0,0x00...	19:	li $v0, 11
	0x00400024	0x24040020	addiu $4,$0,0x00...	20:	addiu $a0, $zero, 32
	0x00400028	0x0000000c	syscall	21:	syscall
	0x0040002c	0x11400002	beq $10,$0,0x000...	24:	beq $t2, $zero, fim
	0x00400030	0x25080001	addiu $8,$8,0x00...	25:	addiu $t0, $t0, 1
	0x00400034	0x0401fff5	bgez $0,0xfffffff5	26:	b while
	0x00400038	0x2402000a	addiu $2,$0,0x00...	29:	li $v0, 10
	0x0040003c	0x0000000c	syscall	30:	syscall

Name	Number	Value
$zero	0	0x00000000
$at	1	0x10010000
$v0	2	0x00000000
$v1	3	0x00000000
$a0	4	0x00000000
$a1	5	0x00000000
$a2	6	0x00000000
$a3	7	0x00000000
$t0	8	0x00000000
$t1	9	0x10010000
$t2	10	0x00000000
$t3	11	0x10010000
$t4	12	0x00000000
$t5	13	0x00000000
$t6	14	0x00000000
$t7	15	0x00000000
$s0	16	0x00000000
$s1	17	0x00000000
$s2	18	0x00000000
$s3	19	0x00000000
$s4	20	0x00000000
$s5	21	0x00000000
$s6	22	0x00000000
$s7	23	0x00000000
$t8	24	0x00000000
$t9	25	0x00000000
$k0	26	0x00000000
$k1	27	0x00000000
$gp	28	0x10008000
$sp	29	0x7fffeffc
$fp	30	0x00000000
$ra	31	0x00000000
pc		0x00400010
hi		0x00000000
lo		0x00000000

Passo 6

Text Segment

Bkpt	Address	Code	Basic		Source
	0x00400000	0x00004021	addu $8,$0,$0	8:	move $t0, $zero
	0x00400004	0x3c011001	lui $1,0x00001001	9:	la $t1, vetor
	0x00400008	0x34290000	ori $9,$1,0x0000...		
	0x0040000c	0x01285821	addu $11,$9,$8	11:	addu $t3, $t1, $t0
	0x00400010	0x816a0000	lb $10,0x0000000...	12:	lb $t2, 0($t3)
	0x00400014	0x24020001	addiu $2,$0,0x00...	15:	li $v0, 1
	0x00400018	0x000a2021	addu $4,$0,$10	16:	move $a0, $t2
	0x0040001c	0x0000000c	syscall	17:	syscall
	0x00400020	0x2402000b	addiu $2,$0,0x00...	19:	li $v0, 11
	0x00400024	0x24040020	addiu $4,$0,0x00...	20:	addiu $a0, $zero, 32
	0x00400028	0x0000000c	syscall	21:	syscall
	0x0040002c	0x11400002	beq $10,$0,0x000...	24:	beq $t2, $zero, fim
	0x00400030	0x25080001	addiu $8,$8,0x00...	25:	addiu $t0, $t0, 1
	0x00400034	0x0401fff5	bgez $0,0xfffffff5	26:	b while
	0x00400038	0x2402000a	addiu $2,$0,0x00...	29:	li $v0, 10
	0x0040003c	0x0000000c	syscall	30:	syscall

Name	Number	Value
$zero	0	0x00000000
$at	1	0x10010000
$v0	2	0x00000000
$v1	3	0x00000000
$a0	4	0x00000000
$a1	5	0x00000000
$a2	6	0x00000000
$a3	7	0x00000000
$t0	8	0x00000000
$t1	9	0x10010000
$t2	10	0x0000000a
$t3	11	0x10010000
$t4	12	0x00000000
$t5	13	0x00000000
$t6	14	0x00000000
$t7	15	0x00000000
$s0	16	0x00000000
$s1	17	0x00000000
$s2	18	0x00000000
$s3	19	0x00000000
$s4	20	0x00000000
$s5	21	0x00000000
$s6	22	0x00000000
$s7	23	0x00000000
$t8	24	0x00000000
$t9	25	0x00000000
$k0	26	0x00000000
$k1	27	0x00000000
$gp	28	0x10008000
$sp	29	0x7fffeffc
$fp	30	0x00000000
$ra	31	0x00000000
pc		0x00400014
hi		0x00000000
lo		0x00000000

Passo 7

Text Segment

Bkpt	Address	Code	Basic	Source
	0x00400000	0x00004021	addu $8,$0,$0	8: move $t0, $zero
	0x00400004	0x3c011001	lui $1,0x00001001	9: la $t1, vetor
	0x00400008	0x34290000	ori $9,$1,0x0000...	
	0x0040000c	0x01285821	addu $11,$9,$8	11: addu $t3, $t1, $t0
	0x00400010	0x816a0000	lb $10,0x0000000...	12: lb $t2, 0($t3)
	0x00400014	0x24020001	addiu $2,$0,0x00...	15: li $v0, 1
	0x00400018	0x000a2021	addu $4,$0,$10	16: move $a0, $t2
	0x0040001c	0x0000000c	syscall	17: syscall
	0x00400020	0x2402000b	addiu $2,$0,0x00...	19: li $v0, 11
	0x00400024	0x24040020	addiu $4,$0,0x00...	20: addiu $a0, $zero, 32
	0x00400028	0x0000000c	syscall	21: syscall
	0x0040002c	0x11400002	beq $10,$0,0x000...	24: beq $t2, $zero, fim
	0x00400030	0x25080001	addiu $8,$8,0x00...	25: addiu $t0, $t0, 1
	0x00400034	0x0401fff5	bgez $0,0xfffffff5	26: b while
	0x00400038	0x2402000a	addiu $2,$0,0x00...	29: li $v0, 10
	0x0040003c	0x0000000c	syscall	30: syscall

Name	Number	Value
$zero	0	0x00000000
$at	1	0x10010000
$v0	2	0x00000001
$v1	3	0x00000000
$a0	4	0x00000000
$a1	5	0x00000000
$a2	6	0x00000000
$a3	7	0x00000000
$t0	8	0x00000000
$t1	9	0x10010000
$t2	10	0x0000000a
$t3	11	0x10010000
$t4	12	0x00000000
$t5	13	0x00000000
$t6	14	0x00000000
$t7	15	0x00000000
$s0	16	0x00000000
$s1	17	0x00000000
$s2	18	0x00000000
$s3	19	0x00000000
$s4	20	0x00000000
$s5	21	0x00000000
$s6	22	0x00000000
$s7	23	0x00000000
$t8	24	0x00000000
$t9	25	0x00000000
$k0	26	0x00000000
$k1	27	0x00000000
$gp	28	0x10008000
$sp	29	0x7fffeffc
$fp	30	0x00000000
$ra	31	0x00000000
pc		0x00400018
hi		0x00000000
lo		0x00000000

Passo 8

Text Segment

Bkpt	Address	Code	Basic		Source
	0x00400000	0x00004021	addu $8,$0,$0	8:	move $t0, $zero
	0x00400004	0x3c011001	lui $1,0x00001001	9:	la $t1, vetor
	0x00400008	0x34290000	ori $9,$1,0x0000...		
	0x0040000c	0x01285821	addu $11,$9,$8	11:	addu $t3, $t1, $t0
	0x00400010	0x816a0000	lb $10,0x0000000...	12:	lb $t2, 0($t3)
	0x00400014	0x24020001	addiu $2,$0,0x00...	15:	li $v0, 1
	0x00400018	0x000a2021	addu $4,$0,$10	16:	move $a0, $t2
	0x0040001c	0x0000000c	syscall	17:	syscall
	0x00400020	0x2402000b	addiu $2,$0,0x00...	19:	li $v0, 11
	0x00400024	0x24040020	addiu $4,$0,0x00...	20:	addiu $a0, $zero, 32
	0x00400028	0x0000000c	syscall	21:	syscall
	0x0040002c	0x11400002	beq $10,$0,0x000...	24:	beq $t2, $zero, fim
	0x00400030	0x25080001	addiu $8,$8,0x00...	25:	addiu $t0, $t0, 1
	0x00400034	0x0401fff5	bgez $0,0xfffffff5	26:	b while
	0x00400038	0x2402000a	addiu $2,$0,0x00...	29:	li $v0, 10
	0x0040003c	0x0000000c	syscall	30:	syscall

Name	Number	Value
$zero	0	0x00000000
$at	1	0x10010000
$v0	2	0x00000001
$v1	3	0x00000000
$a0	4	0x0000000a
$a1	5	0x00000000
$a2	6	0x00000000
$a3	7	0x00000000
$t0	8	0x00000000
$t1	9	0x10010000
$t2	10	0x0000000a
$t3	11	0x10010000
$t4	12	0x00000000
$t5	13	0x00000000
$t6	14	0x00000000
$t7	15	0x00000000
$s0	16	0x00000000
$s1	17	0x00000000
$s2	18	0x00000000
$s3	19	0x00000000
$s4	20	0x00000000
$s5	21	0x00000000
$s6	22	0x00000000
$s7	23	0x00000000
$t8	24	0x00000000
$t9	25	0x00000000
$k0	26	0x00000000
$k1	27	0x00000000
$gp	28	0x10008000
$sp	29	0x7fffeffc
$fp	30	0x00000000
$ra	31	0x00000000
pc		0x0040001c
hi		0x00000000
lo		0x00000000

Imprime 10!

Passo 9

Text Segment

Bkpt	Address	Code	Basic		Source
☐	0x00400000	0x00004021	addu $8,$0,$0	8:	move $t0, $zero
☐	0x00400004	0x3c011001	lui $1,0x00001001	9:	la $t1, vetor
☐	0x00400008	0x34290000	ori $9,$1,0x0000...		
☐	0x0040000c	0x01285821	addu $11,$9,$8	11:	addu $t3, $t1, $t0
☐	0x00400010	0x816a0000	lb $10,0x0000000...	12:	lb $t2, 0($t3)
☐	0x00400014	0x24020001	addiu $2,$0,0x00...	15:	li $v0, 1
☐	0x00400018	0x000a2021	addu $4,$0,$10	16:	move $a0, $t2
☐	0x0040001c	0x0000000c	syscall	17:	syscall
☐	0x00400020	0x2402000b	addiu $2,$0,0x00...	19:	li $v0, 11
☐	0x00400024	0x24040020	addiu $4,$0,0x00...	20:	addiu $a0, $zero, 32
☐	0x00400028	0x0000000c	syscall	21:	syscall
☐	0x0040002c	0x11400002	beq $10,$0,0x000...	24:	beq $t2, $zero, fim
☐	0x00400030	0x25080001	addiu $8,$8,0x00...	25:	addiu $t0, $t0, 1
☐	0x00400034	0x0401fff5	bgez $0,0xfffffff5	26:	b while
☐	0x00400038	0x2402000a	addiu $2,$0,0x00...	29:	li $v0, 10
☐	0x0040003c	0x0000000c	syscall	30:	syscall

Name	Number	Value
$zero	0	0x00000000
$at	1	0x10010000
$v0	2	0x00000001
$v1	3	0x00000000
$a0	4	0x0000000a
$a1	5	0x00000000
$a2	6	0x00000000
$a3	7	0x00000000
$t0	8	0x00000000
$t1	9	0x10010000
$t2	10	0x0000000a
$t3	11	0x10010000
$t4	12	0x00000000
$t5	13	0x00000000
$t6	14	0x00000000
$t7	15	0x00000000
$s0	16	0x00000000
$s1	17	0x00000000
$s2	18	0x00000000
$s3	19	0x00000000
$s4	20	0x00000000
$s5	21	0x00000000
$s6	22	0x00000000
$s7	23	0x00000000
$t8	24	0x00000000
$t9	25	0x00000000
$k0	26	0x00000000
$k1	27	0x00000000
$gp	28	0x10008000
$sp	29	0x7fffeffc
$fp	30	0x00000000
$ra	31	0x00000000
pc		0x00400020
hi		0x00000000
lo		0x00000000

Passo 10

Text Segment

Bkpt	Address	Code	Basic		Source
	0x00400000	0x00004021	addu $8,$0,$0	8:	move $t0, $zero
	0x00400004	0x3c011001	lui $1,0x00001001	9:	la $t1, vetor
	0x00400008	0x34290000	ori $9,$1,0x0000...		
	0x0040000c	0x01285821	addu $11,$9,$8	11:	addu $t3, $t1, $t0
	0x00400010	0x816a0000	lb $10,0x00000000...	12:	lb $t2, 0($t3)
	0x00400014	0x24020001	addiu $2,$0,0x00...	15:	li $v0, 1
	0x00400018	0x000a2021	addu $4,$0,$10	16:	move $a0, $t2
	0x0040001c	0x0000000c	syscall	17:	syscall
	0x00400020	0x2402000b	addiu $2,$0,0x00...	19:	li $v0, 11
	0x00400024	0x24040020	addiu $4,$0,0x00...	20:	addiu $a0, $zero, 32
	0x00400028	0x0000000c	syscall	21:	syscall
	0x0040002c	0x11400002	beq $10,$0,0x000...	24:	beq $t2, $zero, fim
	0x00400030	0x25080001	addiu $8,$8,0x00...	25:	addiu $t0, $t0, 1
	0x00400034	0x0401fff5	bgez $0,0xfffffff5	26:	b while
	0x00400038	0x2402000a	addiu $2,$0,0x00...	29:	li $v0, 10
	0x0040003c	0x0000000c	syscall	30:	syscall

Name	Number	Value
$zero	0	0x00000000
$at	1	0x10010000
$v0	2	0x0000000b
$v1	3	0x00000000
$a0	4	0x0000000a
$a1	5	0x00000000
$a2	6	0x00000000
$a3	7	0x00000000
$t0	8	0x00000000
$t1	9	0x10010000
$t2	10	0x0000000a
$t3	11	0x10010000
$t4	12	0x00000000
$t5	13	0x00000000
$t6	14	0x00000000
$t7	15	0x00000000
$s0	16	0x00000000
$s1	17	0x00000000
$s2	18	0x00000000
$s3	19	0x00000000
$s4	20	0x00000000
$s5	21	0x00000000
$s6	22	0x00000000
$s7	23	0x00000000
$t8	24	0x00000000
$t9	25	0x00000000
$k0	26	0x00000000
$k1	27	0x00000000
$gp	28	0x10008000
$sp	29	0x7fffeffc
$fp	30	0x00000000
$ra	31	0x00000000
pc		0x00400024
hi		0x00000000
lo		0x00000000

Passo 11

Text Segment

Bkpt	Address	Code	Basic		Source
	0x00400000	0x00004021	addu $8,$0,$0	8:	move $t0, $zero
	0x00400004	0x3c011001	lui $1,0x00001001	9:	la $t1, vetor
	0x00400008	0x34290000	ori $9,$1,0x0000...		
	0x0040000c	0x01285821	addu $11,$9,$8	11:	addu $t3, $t1, $t0
	0x00400010	0x816a0000	lb $10,0x0000000...	12:	lb $t2, 0($t3)
	0x00400014	0x24020001	addiu $2,$0,0x00...	15:	li $v0, 1
	0x00400018	0x000a2021	addu $4,$0,$10	16:	move $a0, $t2
	0x0040001c	0x0000000c	syscall	17:	syscall
	0x00400020	0x2402000b	addiu $2,$0,0x00...	19:	li $v0, 11
	0x00400024	0x24040020	addiu $4,$0,0x00...	20:	addiu $a0, $zero, 32
	0x00400028	0x0000000c	syscall	21:	syscall
	0x0040002c	0x11400002	beq $10,$0,0x000...	24:	beq $t2, $zero, fim
	0x00400030	0x25080001	addiu $8,$8,0x00...	25:	addiu $t0, $t0, 1
	0x00400034	0x0401fff5	bgez $0,0xfffffff5	26:	b while
	0x00400038	0x2402000a	addiu $2,$0,0x00...	29:	li $v0, 10
	0x0040003c	0x0000000c	syscall	30:	syscall

Name	Number	Value
$zero	0	0x00000000
$at	1	0x10010000
$v0	2	0x0000000b
$v1	3	0x00000000
$a0	4	0x00000020
$a1	5	0x00000000
$a2	6	0x00000000
$a3	7	0x00000000
$t0	8	0x00000000
$t1	9	0x10010000
$t2	10	0x0000000a
$t3	11	0x10010000
$t4	12	0x00000000
$t5	13	0x00000000
$t6	14	0x00000000
$t7	15	0x00000000
$s0	16	0x00000000
$s1	17	0x00000000
$s2	18	0x00000000
$s3	19	0x00000000
$s4	20	0x00000000
$s5	21	0x00000000
$s6	22	0x00000000
$s7	23	0x00000000
$t8	24	0x00000000
$t9	25	0x00000000
$k0	26	0x00000000
$k1	27	0x00000000
$gp	28	0x10008000
$sp	29	0x7fffeffc
$fp	30	0x00000000
$ra	31	0x00000000
pc		0x00400028
hi		0x00000000
lo		0x00000000

Passo 12

Text Segment

Bkpt	Address	Code	Basic		Source
	0x00400000	0x00004021	addu $8,$0,$0	8:	move $t0, $zero
	0x00400004	0x3c011001	lui $1,0x00001001	9:	la $t1, vetor
	0x00400008	0x34290000	ori $9,$1,0x0000...		
	0x0040000c	0x01285821	addu $11,$9,$8	11:	addu $t3, $t1, $t0
	0x00400010	0x816a0000	lb $10,0x0000000...	12:	lb $t2, 0($t3)
	0x00400014	0x24020001	addiu $2,$0,0x00...	15:	li $v0, 1
	0x00400018	0x000a2021	addu $4,$0,$10	16:	move $a0, $t2
	0x0040001c	0x0000000c	syscall	17:	syscall
	0x00400020	0x2402000b	addiu $2,$0,0x00...	19:	li $v0, 11
	0x00400024	0x24040020	addiu $4,$0,0x00...	20:	addiu $a0, $zero, 32
	0x00400028	0x0000000c	syscall	21:	syscall
	0x0040002c	0x11400002	beq $10,$0,0x000...	24:	beq $t2, $zero, fim
	0x00400030	0x25080001	addiu $8,$8,0x00...	25:	addiu $t0, $t0, 1
	0x00400034	0x0401fff5	bgez $0,0xfffffff5	26:	b while
	0x00400038	0x2402000a	addiu $2,$0,0x00...	29:	li $v0, 10
	0x0040003c	0x0000000c	syscall	30:	syscall

Name	Number	Value
$zero	0	0x00000000
$at	1	0x10010000
$v0	2	0x0000000b
$v1	3	0x00000000
$a0	4	0x00000020
$a1	5	0x00000000
$a2	6	0x00000000
$a3	7	0x00000000
$t0	8	0x00000000
$t1	9	0x10010000
$t2	10	0x0000000a
$t3	11	0x10010000
$t4	12	0x00000000
$t5	13	0x00000000
$t6	14	0x00000000
$t7	15	0x00000000
$s0	16	0x00000000
$s1	17	0x00000000
$s2	18	0x00000000
$s3	19	0x00000000
$s4	20	0x00000000
$s5	21	0x00000000
$s6	22	0x00000000
$s7	23	0x00000000
$t8	24	0x00000000
$t9	25	0x00000000
$k0	26	0x00000000
$k1	27	0x00000000
$gp	28	0x10008000
$sp	29	0x7fffeffc
$fp	30	0x00000000
$ra	31	0x00000000
pc		0x0040002c
hi		0x00000000
lo		0x00000000

Passo 13

Text Segment

Bkpt	Address	Code	Basic		Source
	0x00400000	0x00004021	addu $8,$0,$0	8:	move $t0, $zero
	0x00400004	0x3c011001	lui $1,0x00001001	9:	la $t1, vetor
	0x00400008	0x34290000	ori $9,$1,0x0000...		
	0x0040000c	0x01285821	addu $11,$9,$8	11:	addu $t3, $t1, $t0
	0x00400010	0x816a0000	lb $10,0x0000000...	12:	lb $t2, 0($t3)
	0x00400014	0x24020001	addiu $2,$0,0x00...	15:	li $v0, 1
	0x00400018	0x000a2021	addu $4,$0,$10	16:	move $a0, $t2
	0x0040001c	0x0000000c	syscall	17:	syscall
	0x00400020	0x2402000b	addiu $2,$0,0x00...	19:	li $v0, 11
	0x00400024	0x24040020	addiu $4,$0,0x00...	20:	addiu $a0, $zero, 32
	0x00400028	0x0000000c	syscall	21:	syscall
	0x0040002c	0x11400002	beq $10,$0,0x000...	24:	beq $t2, $zero, fim
	0x00400030	0x25080001	addiu $8,$8,0x00...	25:	addiu $t0, $t0, 1
	0x00400034	0x0401fff5	bgez $0,0xfffffff5	26:	b while
	0x00400038	0x2402000a	addiu $2,$0,0x00...	29:	li $v0, 10
	0x0040003c	0x0000000c	syscall	30:	syscall

Name	Number	Value
$zero	0	0x00000000
$at	1	0x10010000
$v0	2	0x0000000b
$v1	3	0x00000000
$a0	4	0x00000020
$a1	5	0x00000000
$a2	6	0x00000000
$a3	7	0x00000000
$t0	8	0x00000000
$t1	9	0x10010000
$t2	10	0x0000000a
$t3	11	0x10010000
$t4	12	0x00000000
$t5	13	0x00000000
$t6	14	0x00000000
$t7	15	0x00000000
$s0	16	0x00000000
$s1	17	0x00000000
$s2	18	0x00000000
$s3	19	0x00000000
$s4	20	0x00000000
$s5	21	0x00000000
$s6	22	0x00000000
$s7	23	0x00000000
$t8	24	0x00000000
$t9	25	0x00000000
$k0	26	0x00000000
$k1	27	0x00000000
$gp	28	0x10008000
$sp	29	0x7fffeffc
$fp	30	0x00000000
$ra	31	0x00000000
pc		0x00400030
hi		0x00000000
lo		0x00000000

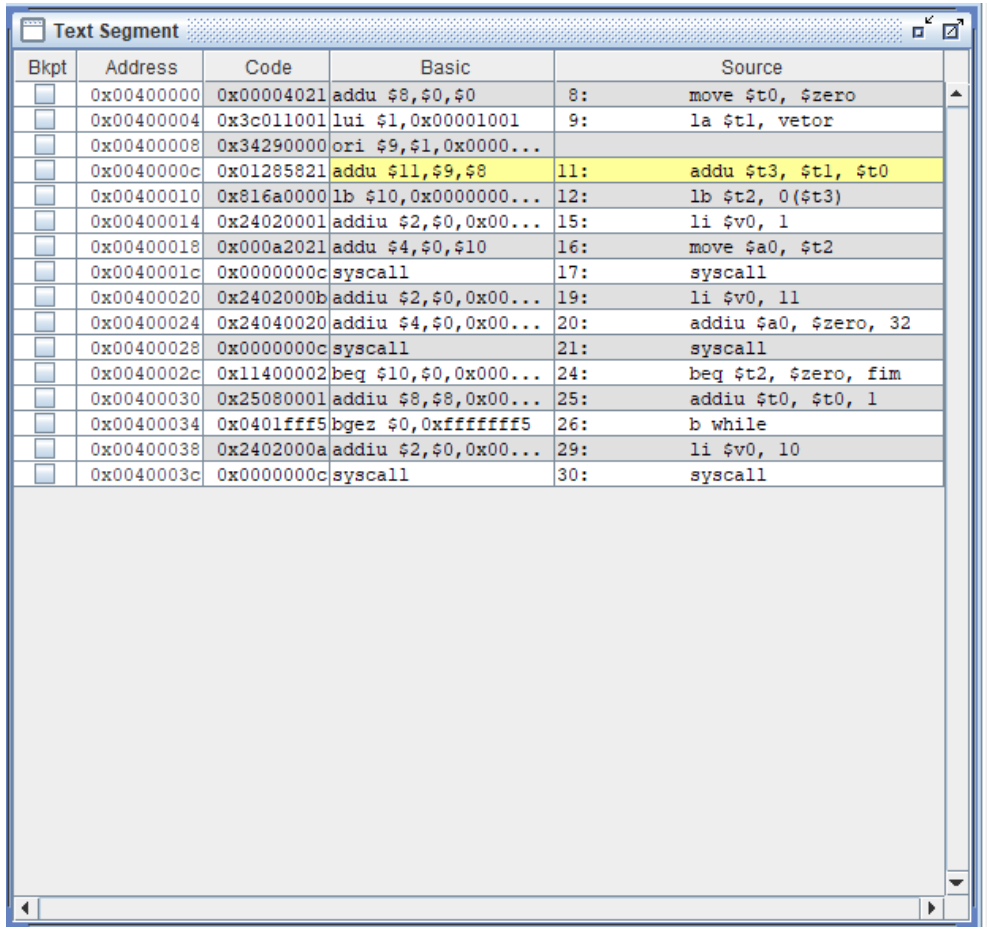
Passo 14

Text Segment

Bkpt	Address	Code	Basic		Source
	0x00400000	0x00004021	addu $8,$0,$0	8:	move $t0, $zero
	0x00400004	0x3c011001	lui $1,0x00001001	9:	la $t1, vetor
	0x00400008	0x34290000	ori $9,$1,0x0000...		
	0x0040000c	0x01285821	addu $11,$9,$8	11:	addu $t3, $t1, $t0
	0x00400010	0x816a0000	lb $10,0x0000000...	12:	lb $t2, 0($t3)
	0x00400014	0x24020001	addiu $2,$0,0x00...	15:	li $v0, 1
	0x00400018	0x000a2021	addu $4,$0,$10	16:	move $a0, $t2
	0x0040001c	0x0000000c	syscall	17:	syscall
	0x00400020	0x2402000b	addiu $2,$0,0x00...	19:	li $v0, 11
	0x00400024	0x24040020	addiu $4,$0,0x00...	20:	addiu $a0, $zero, 32
	0x00400028	0x0000000c	syscall	21:	syscall
	0x0040002c	0x11400002	beq $10,$0,0x000...	24:	beq $t2, $zero, fim
	0x00400030	0x25080001	addiu $8,$8,0x00...	25:	addiu $t0, $t0, 1
	0x00400034	0x0401fff5	bgez $0,0xfffffff5	26:	b while
	0x00400038	0x2402000a	addiu $2,$0,0x00...	29:	li $v0, 10
	0x0040003c	0x0000000c	syscall	30:	syscall

Name	Number	Value
$zero	0	0x00000000
$at	1	0x10010000
$v0	2	0x0000000b
$v1	3	0x00000000
$a0	4	0x00000020
$a1	5	0x00000000
$a2	6	0x00000000
$a3	7	0x00000000
$t0	8	0x00000001
$t1	9	0x10010000
$t2	10	0x0000000a
$t3	11	0x10010000
$t4	12	0x00000000
$t5	13	0x00000000
$t6	14	0x00000000
$t7	15	0x00000000
$s0	16	0x00000000
$s1	17	0x00000000
$s2	18	0x00000000
$s3	19	0x00000000
$s4	20	0x00000000
$s5	21	0x00000000
$s6	22	0x00000000
$s7	23	0x00000000
$t8	24	0x00000000
$t9	25	0x00000000
$k0	26	0x00000000
$k1	27	0x00000000
$gp	28	0x10008000
$sp	29	0x7fffeffc
$fp	30	0x00000000
$ra	31	0x00000000
pc		0x00400034
hi		0x00000000
lo		0x00000000

Passo 15

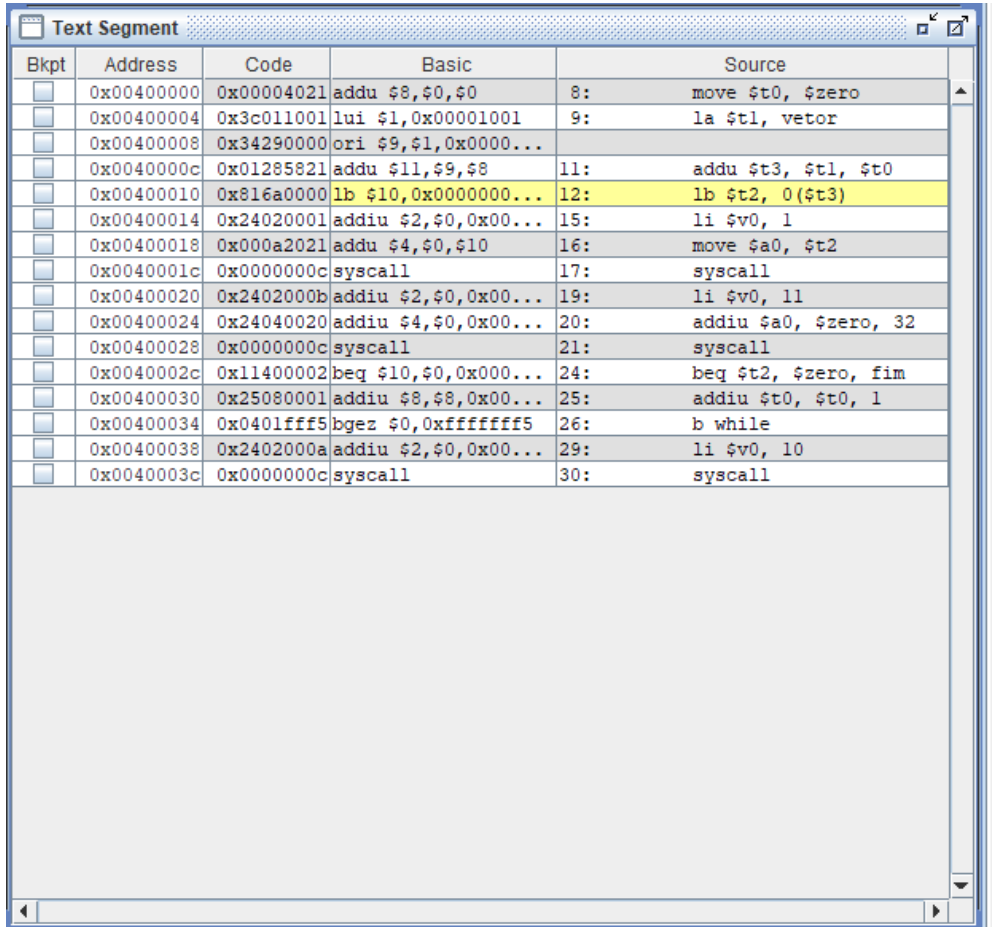


Text Segment

Bkpt	Address	Code	Basic	Source
	0x00400000	0x00004021	addu $8,$0,$0	8: move $t0, $zero
	0x00400004	0x3c011001	lui $1,0x00001001	9: la $t1, vetor
	0x00400008	0x34290000	ori $9,$1,0x0000...	
	0x0040000c	0x01285821	addu $11,$9,$8	11: addu $t3, $t1, $t0
	0x00400010	0x816a0000	lb $10,0x0000000...	12: lb $t2, 0($t3)
	0x00400014	0x24020001	addiu $2,$0,0x00...	15: li $v0, 1
	0x00400018	0x000a2021	addu $4,$0,$10	16: move $a0, $t2
	0x0040001c	0x0000000c	syscall	17: syscall
	0x00400020	0x2402000b	addiu $2,$0,0x00...	19: li $v0, 11
	0x00400024	0x24040020	addiu $4,$0,0x00...	20: addiu $a0, $zero, 32
	0x00400028	0x0000000c	syscall	21: syscall
	0x0040002c	0x11400002	beq $10,$0,0x000...	24: beq $t2, $zero, fim
	0x00400030	0x25080001	addiu $8,$8,0x00...	25: addiu $t0, $t0, 1
	0x00400034	0x0401fff5	bgez $0,0xfffffff5	26: b while
	0x00400038	0x2402000a	addiu $2,$0,0x00...	29: li $v0, 10
	0x0040003c	0x0000000c	syscall	30: syscall

Name	Number	Value
$zero	0	0x00000000
$at	1	0x10010000
$v0	2	0x0000000b
$v1	3	0x00000000
$a0	4	0x00000020
$a1	5	0x00000000
$a2	6	0x00000000
$a3	7	0x00000000
$t0	8	0x00000001
$t1	9	0x10010000
$t2	10	0x0000000a
$t3	11	0x10010000
$t4	12	0x00000000
$t5	13	0x00000000
$t6	14	0x00000000
$t7	15	0x00000000
$s0	16	0x00000000
$s1	17	0x00000000
$s2	18	0x00000000
$s3	19	0x00000000
$s4	20	0x00000000
$s5	21	0x00000000
$s6	22	0x00000000
$s7	23	0x00000000
$t8	24	0x00000000
$t9	25	0x00000000
$k0	26	0x00000000
$k1	27	0x00000000
$gp	28	0x10008000
$sp	29	0x7fffeffc
$fp	30	0x00000000
$ra	31	0x00000000
pc		0x0040000c
hi		0x00000000
lo		0x00000000

Passo 16



Text Segment

Bkpt	Address	Code	Basic		Source
	0x00400000	0x00004021	addu $8,$0,$0	8:	move $t0, $zero
	0x00400004	0x3c011001	lui $1,0x00001001	9:	la $t1, vetor
	0x00400008	0x34290000	ori $9,$1,0x0000...		
	0x0040000c	0x01285821	addu $11,$9,$8	11:	addu $t3, $t1, $t0
	0x00400010	0x816a0000	lb $10,0x00000000...	12:	lb $t2, 0($t3)
	0x00400014	0x24020001	addiu $2,$0,0x00...	15:	li $v0, 1
	0x00400018	0x000a2021	addu $4,$0,$10	16:	move $a0, $t2
	0x0040001c	0x0000000c	syscall	17:	syscall
	0x00400020	0x2402000b	addiu $2,$0,0x00...	19:	li $v0, 11
	0x00400024	0x24040020	addiu $4,$0,0x00...	20:	addiu $a0, $zero, 32
	0x00400028	0x0000000c	syscall	21:	syscall
	0x0040002c	0x11400002	beq $10,$0,0x000...	24:	beq $t2, $zero, fim
	0x00400030	0x25080001	addiu $8,$8,0x00...	25:	addiu $t0, $t0, 1
	0x00400034	0x0401fff5	bgez $0,0xfffffff5	26:	b while
	0x00400038	0x2402000a	addiu $2,$0,0x00...	29:	li $v0, 10
	0x0040003c	0x0000000c	syscall	30:	syscall

Name	Number	Value
$zero	0	0x00000000
$at	1	0x10010000
$v0	2	0x0000000b
$v1	3	0x00000000
$a0	4	0x00000020
$a1	5	0x00000000
$a2	6	0x00000000
$a3	7	0x00000000
$t0	8	0x00000001
$t1	9	0x10010000
$t2	10	0x0000000a
$t3	11	0x10010001
$t4	12	0x00000000
$t5	13	0x00000000
$t6	14	0x00000000
$t7	15	0x00000000
$s0	16	0x00000000
$s1	17	0x00000000
$s2	18	0x00000000
$s3	19	0x00000000
$s4	20	0x00000000
$s5	21	0x00000000
$s6	22	0x00000000
$s7	23	0x00000000
$t8	24	0x00000000
$t9	25	0x00000000
$k0	26	0x00000000
$k1	27	0x00000000
$gp	28	0x10008000
$sp	29	0x7fffeffc
$fp	30	0x00000000
$ra	31	0x00000000
pc		0x00400010
hi		0x00000000
lo		0x00000000

Passo 17

Text Segment

Bkpt	Address	Code	Basic		Source
	0x00400000	0x00004021	addu $8,$0,$0	8:	move $t0, $zero
	0x00400004	0x3c011001	lui $1,0x00001001	9:	la $t1, vetor
	0x00400008	0x34290000	ori $9,$1,0x0000...		
	0x0040000c	0x01285821	addu $11,$9,$8	11:	addu $t3, $t1, $t0
	0x00400010	0x816a0000	lb $10,0x0000000...	12:	lb $t2, 0($t3)
	0x00400014	0x24020001	addiu $2,$0,0x00...	15:	li $v0, 1
	0x00400018	0x000a2021	addu $4,$0,$10	16:	move $a0, $t2
	0x0040001c	0x0000000c	syscall	17:	syscall
	0x00400020	0x2402000b	addiu $2,$0,0x00...	19:	li $v0, 11
	0x00400024	0x24040020	addiu $4,$0,0x00...	20:	addiu $a0, $zero, 32
	0x00400028	0x0000000c	syscall	21:	syscall
	0x0040002c	0x11400002	beq $10,$0,0x000...	24:	beq $t2, $zero, fim
	0x00400030	0x25080001	addiu $8,$8,0x00...	25:	addiu $t0, $t0, 1
	0x00400034	0x0401fff5	bgez $0,0xfffffff5	26:	b while
	0x00400038	0x2402000a	addiu $2,$0,0x00...	29:	li $v0, 10
	0x0040003c	0x0000000c	syscall	30:	syscall

Name	Number	Value
$zero	0	0x00000000
$at	1	0x10010000
$v0	2	0x0000000b
$v1	3	0x00000000
$a0	4	0x00000020
$a1	5	0x00000000
$a2	6	0x00000000
$a3	7	0x00000000
$t0	8	0x00000001
$t1	9	0x10010000
$t2	10	0x00000014
$t3	11	0x10010001
$t4	12	0x00000000
$t5	13	0x00000000
$t6	14	0x00000000
$t7	15	0x00000000
$s0	16	0x00000000
$s1	17	0x00000000
$s2	18	0x00000000
$s3	19	0x00000000
$s4	20	0x00000000
$s5	21	0x00000000
$s6	22	0x00000000
$s7	23	0x00000000
$t8	24	0x00000000
$t9	25	0x00000000
$k0	26	0x00000000
$k1	27	0x00000000
$gp	28	0x10008000
$sp	29	0x7fffeffc
$fp	30	0x00000000
$ra	31	0x00000000
pc		0x00400014
hi		0x00000000
lo		0x00000000

Passo 18

Text Segment

Bkpt	Address	Code	Basic		Source
	0x00400000	0x00004021	addu $8,$0,$0	8:	move $t0, $zero
	0x00400004	0x3c011001	lui $1,0x00001001	9:	la $t1, vetor
	0x00400008	0x34290000	ori $9,$1,0x0000...		
	0x0040000c	0x01285821	addu $11,$9,$8	11:	addu $t3, $t1, $t0
	0x00400010	0x816a0000	lb $10,0x0000000...	12:	lb $t2, 0($t3)
	0x00400014	0x24020001	addiu $2,$0,0x00...	15:	li $v0, 1
	0x00400018	0x000a2021	addu $4,$0,$10	16:	move $a0, $t2
	0x0040001c	0x0000000c	syscall	17:	syscall
	0x00400020	0x2402000b	addiu $2,$0,0x00...	19:	li $v0, 11
	0x00400024	0x24040020	addiu $4,$0,0x00...	20:	addiu $a0, $zero, 32
	0x00400028	0x0000000c	syscall	21:	syscall
	0x0040002c	0x11400002	beq $10,$0,0x000...	24:	beq $t2, $zero, fim
	0x00400030	0x25080001	addiu $8,$8,0x00...	25:	addiu $t0, $t0, 1
	0x00400034	0x0401fff5	bgez $0,0xfffffff5	26:	b while
	0x00400038	0x2402000a	addiu $2,$0,0x00...	29:	li $v0, 10
	0x0040003c	0x0000000c	syscall	30:	syscall

Name	Number	Value
$zero	0	0x00000000
$at	1	0x10010000
$v0	2	0x00000001
$v1	3	0x00000000
$a0	4	0x00000020
$a1	5	0x00000000
$a2	6	0x00000000
$a3	7	0x00000000
$t0	8	0x00000001
$t1	9	0x10010000
$t2	10	0x00000014
$t3	11	0x10010001
$t4	12	0x00000000
$t5	13	0x00000000
$t6	14	0x00000000
$t7	15	0x00000000
$s0	16	0x00000000
$s1	17	0x00000000
$s2	18	0x00000000
$s3	19	0x00000000
$s4	20	0x00000000
$s5	21	0x00000000
$s6	22	0x00000000
$s7	23	0x00000000
$t8	24	0x00000000
$t9	25	0x00000000
$k0	26	0x00000000
$k1	27	0x00000000
$gp	28	0x10008000
$sp	29	0x7fffeffc
$fp	30	0x00000000
$ra	31	0x00000000
pc		0x00400018
hi		0x00000000
lo		0x00000000

Passo 19

Text Segment

Bkpt	Address	Code	Basic	Source
	0x00400000	0x00004021	addu $8,$0,$0	8: move $t0, $zero
	0x00400004	0x3c011001	lui $1,0x00001001	9: la $t1, vetor
	0x00400008	0x34290000	ori $9,$1,0x0000...	
	0x0040000c	0x01285821	addu $11,$9,$8	11: addu $t3, $t1, $t0
	0x00400010	0x816a0000	lb $10,0x0000000...	12: lb $t2, 0($t3)
	0x00400014	0x24020001	addiu $2,$0,0x00...	15: li $v0, 1
	0x00400018	0x000a2021	addu $4,$0,$10	16: move $a0, $t2
	0x0040001c	0x0000000c	syscall	17: syscall
	0x00400020	0x2402000b	addiu $2,$0,0x00...	19: li $v0, 11
	0x00400024	0x24040020	addiu $4,$0,0x00...	20: addiu $a0, $zero, 32
	0x00400028	0x0000000c	syscall	21: syscall
	0x0040002c	0x11400002	beq $10,$0,0x000...	24: beq $t2, $zero, fim
	0x00400030	0x25080001	addiu $8,$8,0x00...	25: addiu $t0, $t0, 1
	0x00400034	0x0401fff5	bgez $0,0xfffffff5	26: b while
	0x00400038	0x2402000a	addiu $2,$0,0x00...	29: li $v0, 10
	0x0040003c	0x0000000c	syscall	30: syscall

Name	Number	Value
$zero	0	0x00000000
$at	1	0x10010000
$v0	2	0x00000001
$v1	3	0x00000000
$a0	4	0x00000014
$a1	5	0x00000000
$a2	6	0x00000000
$a3	7	0x00000000
$t0	8	0x00000001
$t1	9	0x10010000
$t2	10	0x00000014
$t3	11	0x10010001
$t4	12	0x00000000
$t5	13	0x00000000
$t6	14	0x00000000
$t7	15	0x00000000
$s0	16	0x00000000
$s1	17	0x00000000
$s2	18	0x00000000
$s3	19	0x00000000
$s4	20	0x00000000
$s5	21	0x00000000
$s6	22	0x00000000
$s7	23	0x00000000
$t8	24	0x00000000
$t9	25	0x00000000
$k0	26	0x00000000
$k1	27	0x00000000
$gp	28	0x10008000
$sp	29	0x7fffeffc
$fp	30	0x00000000
$ra	31	0x00000000
pc		0x0040001c
hi		0x00000000
lo		0x00000000

Imprime 20!

Passo 20

Text Segment

Bkpt	Address	Code	Basic	Source
	0x00400000	0x00004021	addu $8,$0,$0	8: move $t0, $zero
	0x00400004	0x3c011001	lui $1,0x00001001	9: la $t1, vetor
	0x00400008	0x34290000	ori $9,$1,0x0000...	
	0x0040000c	0x01285821	addu $11,$9,$8	11: addu $t3, $t1, $t0
	0x00400010	0x816a0000	lb $10,0x0000000...	12: lb $t2, 0($t3)
	0x00400014	0x24020001	addiu $2,$0,0x00...	15: li $v0, 1
	0x00400018	0x000a2021	addu $4,$0,$10	16: move $a0, $t2
	0x0040001c	0x0000000c	syscall	17: syscall
	0x00400020	0x2402000b	addiu $2,$0,0x00...	19: li $v0, 11
	0x00400024	0x24040020	addiu $4,$0,0x00...	20: addiu $a0, $zero, 32
	0x00400028	0x0000000c	syscall	21: syscall
	0x0040002c	0x11400002	beq $10,$0,0x000...	24: beq $t2, $zero, fim
	0x00400030	0x25080001	addiu $8,$8,0x00...	25: addiu $t0, $t0, 1
	0x00400034	0x0401fff5	bgez $0,0xfffffff5	26: b while
	0x00400038	0x2402000a	addiu $2,$0,0x00...	29: li $v0, 10
	0x0040003c	0x0000000c	syscall	30: syscall

Name	Number	Value
$zero	0	0x00000000
$at	1	0x10010000
$v0	2	0x00000001
$v1	3	0x00000000
$a0	4	0x00000014
$a1	5	0x00000000
$a2	6	0x00000000
$a3	7	0x00000000
$t0	8	0x00000001
$t1	9	0x10010000
$t2	10	0x00000014
$t3	11	0x10010001
$t4	12	0x00000000
$t5	13	0x00000000
$t6	14	0x00000000
$t7	15	0x00000000
$s0	16	0x00000000
$s1	17	0x00000000
$s2	18	0x00000000
$s3	19	0x00000000
$s4	20	0x00000000
$s5	21	0x00000000
$s6	22	0x00000000
$s7	23	0x00000000
$t8	24	0x00000000
$t9	25	0x00000000
$k0	26	0x00000000
$k1	27	0x00000000
$gp	28	0x10008000
$sp	29	0x7fffeffc
$fp	30	0x00000000
$ra	31	0x00000000
pc		0x00400020
hi		0x00000000
lo		0x00000000

Passo 21

Text Segment

Bkpt	Address	Code	Basic		Source
	0x00400000	0x00004021	addu $8,$0,$0	8:	move $t0, $zero
	0x00400004	0x3c011001	lui $1,0x00001001	9:	la $t1, vetor
	0x00400008	0x34290000	ori $9,$1,0x0000...		
	0x0040000c	0x01285821	addu $11,$9,$8	11:	addu $t3, $t1, $t0
	0x00400010	0x816a0000	lb $10,0x0000000...	12:	lb $t2, 0($t3)
	0x00400014	0x24020001	addiu $2,$0,0x00...	15:	li $v0, 1
	0x00400018	0x000a2021	addu $4,$0,$10	16:	move $a0, $t2
	0x0040001c	0x0000000c	syscall	17:	syscall
	0x00400020	0x2402000b	addiu $2,$0,0x00...	19:	li $v0, 11
	0x00400024	0x24040020	addiu $4,$0,0x00...	20:	addiu $a0, $zero, 32
	0x00400028	0x0000000c	syscall	21:	syscall
	0x0040002c	0x11400002	beq $10,$0,0x000...	24:	beq $t2, $zero, fim
	0x00400030	0x25080001	addiu $8,$8,0x00...	25:	addiu $t0, $t0, 1
	0x00400034	0x0401fff5	bgez $0,0xfffffff5	26:	b while
	0x00400038	0x2402000a	addiu $2,$0,0x00...	29:	li $v0, 10
	0x0040003c	0x0000000c	syscall	30:	syscall

Name	Number	Value
$zero	0	0x00000000
$at	1	0x10010000
$v0	2	0x0000000b
$v1	3	0x00000000
$a0	4	0x00000014
$a1	5	0x00000000
$a2	6	0x00000000
$a3	7	0x00000000
$t0	8	0x00000001
$t1	9	0x10010000
$t2	10	0x00000014
$t3	11	0x10010001
$t4	12	0x00000000
$t5	13	0x00000000
$t6	14	0x00000000
$t7	15	0x00000000
$s0	16	0x00000000
$s1	17	0x00000000
$s2	18	0x00000000
$s3	19	0x00000000
$s4	20	0x00000000
$s5	21	0x00000000
$s6	22	0x00000000
$s7	23	0x00000000
$t8	24	0x00000000
$t9	25	0x00000000
$k0	26	0x00000000
$k1	27	0x00000000
$gp	28	0x10008000
$sp	29	0x7fffeffc
$fp	30	0x00000000
$ra	31	0x00000000
pc		0x00400024
hi		0x00000000
lo		0x00000000

Passo 22

Text Segment

Bkpt	Address	Code	Basic		Source
	0x00400000	0x00004021	addu $8,$0,$0	8:	move $t0, $zero
	0x00400004	0x3c011001	lui $1,0x00001001	9:	la $t1, vetor
	0x00400008	0x34290000	ori $9,$1,0x0000...		
	0x0040000c	0x01285821	addu $11,$9,$8	11:	addu $t3, $t1, $t0
	0x00400010	0x816a0000	lb $10,0x00000000...	12:	lb $t2, 0($t3)
	0x00400014	0x24020001	addiu $2,$0,0x00...	15:	li $v0, 1
	0x00400018	0x000a2021	addu $4,$0,$10	16:	move $a0, $t2
	0x0040001c	0x0000000c	syscall	17:	syscall
	0x00400020	0x2402000b	addiu $2,$0,0x00...	19:	li $v0, 11
	0x00400024	0x24040020	addiu $4,$0,0x00...	20:	addiu $a0, $zero, 32
	0x00400028	0x0000000c	syscall	21:	syscall
	0x0040002c	0x11400002	beq $10,$0,0x000...	24:	beq $t2, $zero, fim
	0x00400030	0x25080001	addiu $8,$8,0x00...	25:	addiu $t0, $t0, 1
	0x00400034	0x0401fff5	bgez $0,0xfffffff5	26:	b while
	0x00400038	0x2402000a	addiu $2,$0,0x00...	29:	li $v0, 10
	0x0040003c	0x0000000c	syscall	30:	syscall

Name	Number	Value
$zero	0	0x00000000
$at	1	0x10010000
$v0	2	0x0000000b
$v1	3	0x00000000
$a0	4	0x00000020
$a1	5	0x00000000
$a2	6	0x00000000
$a3	7	0x00000000
$t0	8	0x00000001
$t1	9	0x10010000
$t2	10	0x00000014
$t3	11	0x10010001
$t4	12	0x00000000
$t5	13	0x00000000
$t6	14	0x00000000
$t7	15	0x00000000
$s0	16	0x00000000
$s1	17	0x00000000
$s2	18	0x00000000
$s3	19	0x00000000
$s4	20	0x00000000
$s5	21	0x00000000
$s6	22	0x00000000
$s7	23	0x00000000
$t8	24	0x00000000
$t9	25	0x00000000
$k0	26	0x00000000
$k1	27	0x00000000
$gp	28	0x10008000
$sp	29	0x7fffeffc
$fp	30	0x00000000
$ra	31	0x00000000
pc		0x00400028
hi		0x00000000
lo		0x00000000

Passo 23

Text Segment

Bkpt	Address	Code	Basic		Source
	0x00400000	0x00004021	addu $8,$0,$0	8:	move $t0, $zero
	0x00400004	0x3c011001	lui $1,0x00001001	9:	la $t1, vetor
	0x00400008	0x34290000	ori $9,$1,0x0000...		
	0x0040000c	0x01285821	addu $11,$9,$8	11:	addu $t3, $t1, $t0
	0x00400010	0x816a0000	lb $10,0x0000000...	12:	lb $t2, 0($t3)
	0x00400014	0x24020001	addiu $2,$0,0x00...	15:	li $v0, 1
	0x00400018	0x000a2021	addu $4,$0,$10	16:	move $a0, $t2
	0x0040001c	0x0000000c	syscall	17:	syscall
	0x00400020	0x2402000b	addiu $2,$0,0x00...	19:	li $v0, 11
	0x00400024	0x24040020	addiu $4,$0,0x00...	20:	addiu $a0, $zero, 32
	0x00400028	0x0000000c	syscall	21:	syscall
	0x0040002c	0x11400002	beq $10,$0,0x000...	24:	beq $t2, $zero, fim
	0x00400030	0x25080001	addiu $8,$8,0x00...	25:	addiu $t0, $t0, 1
	0x00400034	0x0401fff5	bgez $0,0xfffffff5	26:	b while
	0x00400038	0x2402000a	addiu $2,$0,0x00...	29:	li $v0, 10
	0x0040003c	0x0000000c	syscall	30:	syscall

Name	Number	Value
$zero	0	0x00000000
$at	1	0x10010000
$v0	2	0x0000000b
$v1	3	0x00000000
$a0	4	0x00000020
$a1	5	0x00000000
$a2	6	0x00000000
$a3	7	0x00000000
$t0	8	0x00000001
$t1	9	0x10010000
$t2	10	0x00000014
$t3	11	0x10010001
$t4	12	0x00000000
$t5	13	0x00000000
$t6	14	0x00000000
$t7	15	0x00000000
$s0	16	0x00000000
$s1	17	0x00000000
$s2	18	0x00000000
$s3	19	0x00000000
$s4	20	0x00000000
$s5	21	0x00000000
$s6	22	0x00000000
$s7	23	0x00000000
$t8	24	0x00000000
$t9	25	0x00000000
$k0	26	0x00000000
$k1	27	0x00000000
$gp	28	0x10008000
$sp	29	0x7fffeffc
$fp	30	0x00000000
$ra	31	0x00000000
pc		0x0040002c
hi		0x00000000
lo		0x00000000

Passo 24

Text Segment

Bkpt	Address	Code	Basic		Source
	0x00400000	0x00004021	addu $8,$0,$0	8:	move $t0, $zero
	0x00400004	0x3c011001	lui $1,0x00001001	9:	la $t1, vetor
	0x00400008	0x34290000	ori $9,$1,0x0000...		
	0x0040000c	0x01285821	addu $11,$9,$8	11:	addu $t3, $t1, $t0
	0x00400010	0x816a0000	lb $10,0x0000000...	12:	lb $t2, 0($t3)
	0x00400014	0x24020001	addiu $2,$0,0x00...	15:	li $v0, 1
	0x00400018	0x000a2021	addu $4,$0,$10	16:	move $a0, $t2
	0x0040001c	0x0000000c	syscall	17:	syscall
	0x00400020	0x2402000b	addiu $2,$0,0x00...	19:	li $v0, 11
	0x00400024	0x24040020	addiu $4,$0,0x00...	20:	addiu $a0, $zero, 32
	0x00400028	0x0000000c	syscall	21:	syscall
	0x0040002c	0x11400002	beq $10,$0,0x000...	24:	beq $t2, $zero, fim
	0x00400030	0x25080001	addiu $8,$8,0x00...	25:	addiu $t0, $t0, 1
	0x00400034	0x0401fff5	bgez $0,0xfffffff5	26:	b while
	0x00400038	0x2402000a	addiu $2,$0,0x00...	29:	li $v0, 10
	0x0040003c	0x0000000c	syscall	30:	syscall

Name	Number	Value
$zero	0	0x00000000
$at	1	0x10010000
$v0	2	0x0000000b
$v1	3	0x00000000
$a0	4	0x00000020
$a1	5	0x00000000
$a2	6	0x00000000
$a3	7	0x00000000
$t0	8	0x00000001
$t1	9	0x10010000
$t2	10	0x00000014
$t3	11	0x10010001
$t4	12	0x00000000
$t5	13	0x00000000
$t6	14	0x00000000
$t7	15	0x00000000
$s0	16	0x00000000
$s1	17	0x00000000
$s2	18	0x00000000
$s3	19	0x00000000
$s4	20	0x00000000
$s5	21	0x00000000
$s6	22	0x00000000
$s7	23	0x00000000
$t8	24	0x00000000
$t9	25	0x00000000
$k0	26	0x00000000
$k1	27	0x00000000
$gp	28	0x10008000
$sp	29	0x7fffeffc
$fp	30	0x00000000
$ra	31	0x00000000
pc		0x00400030
hi		0x00000000
lo		0x00000000

Passo 25

Text Segment

Bkpt	Address	Code	Basic		Source
	0x00400000	0x00004021	addu $8,$0,$0	8:	move $t0, $zero
	0x00400004	0x3c011001	lui $1,0x00001001	9:	la $t1, vetor
	0x00400008	0x34290000	ori $9,$1,0x0000...		
	0x0040000c	0x01285821	addu $11,$9,$8	11:	addu $t3, $t1, $t0
	0x00400010	0x816a0000	lb $10,0x0000000...	12:	lb $t2, 0($t3)
	0x00400014	0x24020001	addiu $2,$0,0x00...	15:	li $v0, 1
	0x00400018	0x000a2021	addu $4,$0,$10	16:	move $a0, $t2
	0x0040001c	0x0000000c	syscall	17:	syscall
	0x00400020	0x2402000b	addiu $2,$0,0x00...	19:	li $v0, 11
	0x00400024	0x24040020	addiu $4,$0,0x00...	20:	addiu $a0, $zero, 32
	0x00400028	0x0000000c	syscall	21:	syscall
	0x0040002c	0x11400002	beq $10,$0,0x000...	24:	beq $t2, $zero, fim
	0x00400030	0x25080001	addiu $8,$8,0x00...	25:	addiu $t0, $t0, 1
	0x00400034	0x0401fff5	bgez $0,0xfffffff5	26:	b while
	0x00400038	0x2402000a	addiu $2,$0,0x00...	29:	li $v0, 10
	0x0040003c	0x0000000c	syscall	30:	syscall

Name	Number	Value
$zero	0	0x00000000
$at	1	0x10010000
$v0	2	0x0000000b
$v1	3	0x00000000
$a0	4	0x00000020
$a1	5	0x00000000
$a2	6	0x00000000
$a3	7	0x00000000
$t0	8	0x00000002
$t1	9	0x10010000
$t2	10	0x00000014
$t3	11	0x10010001
$t4	12	0x00000000
$t5	13	0x00000000
$t6	14	0x00000000
$t7	15	0x00000000
$s0	16	0x00000000
$s1	17	0x00000000
$s2	18	0x00000000
$s3	19	0x00000000
$s4	20	0x00000000
$s5	21	0x00000000
$s6	22	0x00000000
$s7	23	0x00000000
$t8	24	0x00000000
$t9	25	0x00000000
$k0	26	0x00000000
$k1	27	0x00000000
$gp	28	0x10008000
$sp	29	0x7fffeffc
$fp	30	0x00000000
$ra	31	0x00000000
pc		0x00400034
hi		0x00000000
lo		0x00000000

Passo 26

Text Segment

Bkpt	Address	Code	Basic		Source
	0x00400000	0x00004021	addu $8,$0,$0	8:	move $t0, $zero
	0x00400004	0x3c011001	lui $1,0x00001001	9:	la $t1, vetor
	0x00400008	0x34290000	ori $9,$1,0x0000...		
	0x0040000c	0x01285821	addu $11,$9,$8	11:	addu $t3, $t1, $t0
	0x00400010	0x816a0000	lb $10,0x0000000...	12:	lb $t2, 0($t3)
	0x00400014	0x24020001	addiu $2,$0,0x00...	15:	li $v0, 1
	0x00400018	0x000a2021	addu $4,$0,$10	16:	move $a0, $t2
	0x0040001c	0x0000000c	syscall	17:	syscall
	0x00400020	0x2402000b	addiu $2,$0,0x00...	19:	li $v0, 11
	0x00400024	0x24040020	addiu $4,$0,0x00...	20:	addiu $a0, $zero, 32
	0x00400028	0x0000000c	syscall	21:	syscall
	0x0040002c	0x11400002	beq $10,$0,0x000...	24:	beq $t2, $zero, fim
	0x00400030	0x25080001	addiu $8,$8,0x00...	25:	addiu $t0, $t0, 1
	0x00400034	0x0401fff5	bgez $0,0xfffffff5	26:	b while
	0x00400038	0x2402000a	addiu $2,$0,0x00...	29:	li $v0, 10
	0x0040003c	0x0000000c	syscall	30:	syscall

Name	Number	Value
$zero	0	0x00000000
$at	1	0x10010000
$v0	2	0x0000000b
$v1	3	0x00000000
$a0	4	0x00000020
$a1	5	0x00000000
$a2	6	0x00000000
$a3	7	0x00000000
$t0	8	0x00000002
$t1	9	0x10010000
$t2	10	0x00000014
$t3	11	0x10010001
$t4	12	0x00000000
$t5	13	0x00000000
$t6	14	0x00000000
$t7	15	0x00000000
$s0	16	0x00000000
$s1	17	0x00000000
$s2	18	0x00000000
$s3	19	0x00000000
$s4	20	0x00000000
$s5	21	0x00000000
$s6	22	0x00000000
$s7	23	0x00000000
$t8	24	0x00000000
$t9	25	0x00000000
$k0	26	0x00000000
$k1	27	0x00000000
$gp	28	0x10008000
$sp	29	0x7fffeffc
$fp	30	0x00000000
$ra	31	0x00000000
pc		0x0040000c
hi		0x00000000
lo		0x00000000

Passo 27

Text Segment

Bkpt	Address	Code	Basic		Source
	0x00400000	0x00004021	addu $8,$0,$0	8:	move $t0, $zero
	0x00400004	0x3c011001	lui $1,0x00001001	9:	la $t1, vetor
	0x00400008	0x34290000	ori $9,$1,0x0000...		
	0x0040000c	0x01285821	addu $11,$9,$8	11:	addu $t3, $t1, $t0
	0x00400010	0x816a0000	lb $10,0x0000000...	12:	lb $t2, 0($t3)
	0x00400014	0x24020001	addiu $2,$0,0x00...	15:	li $v0, 1
	0x00400018	0x000a2021	addu $4,$0,$10	16:	move $a0, $t2
	0x0040001c	0x0000000c	syscall	17:	syscall
	0x00400020	0x2402000b	addiu $2,$0,0x00...	19:	li $v0, 11
	0x00400024	0x24040020	addiu $4,$0,0x00...	20:	addiu $a0, $zero, 32
	0x00400028	0x0000000c	syscall	21:	syscall
	0x0040002c	0x11400002	beq $10,$0,0x000...	24:	beq $t2, $zero, fim
	0x00400030	0x25080001	addiu $8,$8,0x00...	25:	addiu $t0, $t0, 1
	0x00400034	0x0401fff5	bgez $0,0xfffffff5	26:	b while
	0x00400038	0x2402000a	addiu $2,$0,0x00...	29:	li $v0, 10
	0x0040003c	0x0000000c	syscall	30:	syscall

Name	Number	Value
$zero	0	0x00000000
$at	1	0x10010000
$v0	2	0x0000000b
$v1	3	0x00000000
$a0	4	0x00000020
$a1	5	0x00000000
$a2	6	0x00000000
$a3	7	0x00000000
$t0	8	0x00000002
$t1	9	0x10010000
$t2	10	0x00000014
$t3	11	0x10010002
$t4	12	0x00000000
$t5	13	0x00000000
$t6	14	0x00000000
$t7	15	0x00000000
$s0	16	0x00000000
$s1	17	0x00000000
$s2	18	0x00000000
$s3	19	0x00000000
$s4	20	0x00000000
$s5	21	0x00000000
$s6	22	0x00000000
$s7	23	0x00000000
$t8	24	0x00000000
$t9	25	0x00000000
$k0	26	0x00000000
$k1	27	0x00000000
$gp	28	0x10008000
$sp	29	0x7fffeffc
$fp	30	0x00000000
$ra	31	0x00000000
pc		0x00400010
hi		0x00000000
lo		0x00000000

Passo 28

Text Segment

Bkpt	Address	Code	Basic		Source
	0x00400000	0x00004021	addu $8,$0,$0	8:	move $t0, $zero
	0x00400004	0x3c011001	lui $1,0x00001001	9:	la $t1, vetor
	0x00400008	0x34290000	ori $9,$1,0x0000...		
	0x0040000c	0x01285821	addu $11,$9,$8	11:	addu $t3, $t1, $t0
	0x00400010	0x816a0000	lb $10,0x0000000...	12:	lb $t2, 0($t3)
	0x00400014	0x24020001	addiu $2,$0,0x00...	15:	li $v0, 1
	0x00400018	0x000a2021	addu $4,$0,$10	16:	move $a0, $t2
	0x0040001c	0x0000000c	syscall	17:	syscall
	0x00400020	0x2402000b	addiu $2,$0,0x00...	19:	li $v0, 11
	0x00400024	0x24040020	addiu $4,$0,0x00...	20:	addiu $a0, $zero, 32
	0x00400028	0x0000000c	syscall	21:	syscall
	0x0040002c	0x11400002	beq $10,$0,0x000...	24:	beq $t2, $zero, fim
	0x00400030	0x25080001	addiu $8,$8,0x00...	25:	addiu $t0, $t0, 1
	0x00400034	0x0401fff5	bgez $0,0xfffffff5	26:	b while
	0x00400038	0x2402000a	addiu $2,$0,0x00...	29:	li $v0, 10
	0x0040003c	0x0000000c	syscall	30:	syscall

Name	Number	Value
$zero	0	0x00000000
$at	1	0x10010000
$v0	2	0x0000000b
$v1	3	0x00000000
$a0	4	0x00000020
$a1	5	0x00000000
$a2	6	0x00000000
$a3	7	0x00000000
$t0	8	0x00000002
$t1	9	0x10010000
$t2	10	0x0000001e
$t3	11	0x10010002
$t4	12	0x00000000
$t5	13	0x00000000
$t6	14	0x00000000
$t7	15	0x00000000
$s0	16	0x00000000
$s1	17	0x00000000
$s2	18	0x00000000
$s3	19	0x00000000
$s4	20	0x00000000
$s5	21	0x00000000
$s6	22	0x00000000
$s7	23	0x00000000
$t8	24	0x00000000
$t9	25	0x00000000
$k0	26	0x00000000
$k1	27	0x00000000
$gp	28	0x10008000
$sp	29	0x7fffeffc
$fp	30	0x00000000
$ra	31	0x00000000
pc		0x00400014
hi		0x00000000
lo		0x00000000

Passo 29

Text Segment

Bkpt	Address	Code	Basic	Source
	0x00400000	0x00004021	addu $8,$0,$0	8: move $t0, $zero
	0x00400004	0x3c011001	lui $1,0x00001001	9: la $t1, vetor
	0x00400008	0x34290000	ori $9,$1,0x0000...	
	0x0040000c	0x01285821	addu $11,$9,$8	11: addu $t3, $t1, $t0
	0x00400010	0x816a0000	lb $10,0x0000000...	12: lb $t2, 0($t3)
	0x00400014	0x24020001	addiu $2,$0,0x00...	15: li $v0, 1
	0x00400018	0x000a2021	addu $4,$0,$10	16: move $a0, $t2
	0x0040001c	0x0000000c	syscall	17: syscall
	0x00400020	0x2402000b	addiu $2,$0,0x00...	19: li $v0, 11
	0x00400024	0x24040020	addiu $4,$0,0x00...	20: addiu $a0, $zero, 32
	0x00400028	0x0000000c	syscall	21: syscall
	0x0040002c	0x11400002	beq $10,$0,0x000...	24: beq $t2, $zero, fim
	0x00400030	0x25080001	addiu $8,$8,0x00...	25: addiu $t0, $t0, 1
	0x00400034	0x0401fff5	bgez $0,0xfffffff5	26: b while
	0x00400038	0x2402000a	addiu $2,$0,0x00...	29: li $v0, 10
	0x0040003c	0x0000000c	syscall	30: syscall

Name	Number	Value
$zero	0	0x00000000
$at	1	0x10010000
$v0	2	0x00000001
$v1	3	0x00000000
$a0	4	0x00000020
$a1	5	0x00000000
$a2	6	0x00000000
$a3	7	0x00000000
$t0	8	0x00000002
$t1	9	0x10010000
$t2	10	0x0000001e
$t3	11	0x10010002
$t4	12	0x00000000
$t5	13	0x00000000
$t6	14	0x00000000
$t7	15	0x00000000
$s0	16	0x00000000
$s1	17	0x00000000
$s2	18	0x00000000
$s3	19	0x00000000
$s4	20	0x00000000
$s5	21	0x00000000
$s6	22	0x00000000
$s7	23	0x00000000
$t8	24	0x00000000
$t9	25	0x00000000
$k0	26	0x00000000
$k1	27	0x00000000
$gp	28	0x10008000
$sp	29	0x7fffeffc
$fp	30	0x00000000
$ra	31	0x00000000
pc		0x00400018
hi		0x00000000
lo		0x00000000

Passo 30

Text Segment

Bkpt	Address	Code	Basic		Source
	0x00400000	0x00004021	addu $8,$0,$0	8:	move $t0, $zero
	0x00400004	0x3c011001	lui $1,0x00001001	9:	la $t1, vetor
	0x00400008	0x34290000	ori $9,$1,0x0000...		
	0x0040000c	0x01285821	addu $11,$9,$8	11:	addu $t3, $t1, $t0
	0x00400010	0x816a0000	lb $10,0x0000000...	12:	lb $t2, 0($t3)
	0x00400014	0x24020001	addiu $2,$0,0x00...	15:	li $v0, 1
	0x00400018	0x000a2021	addu $4,$0,$10	16:	move $a0, $t2
	0x0040001c	0x0000000c	syscall	17:	syscall
	0x00400020	0x2402000b	addiu $2,$0,0x00...	19:	li $v0, 11
	0x00400024	0x24040020	addiu $4,$0,0x00...	20:	addiu $a0, $zero, 32
	0x00400028	0x0000000c	syscall	21:	syscall
	0x0040002c	0x11400002	beq $10,$0,0x000...	24:	beq $t2, $zero, fim
	0x00400030	0x25080001	addiu $8,$8,0x00...	25:	addiu $t0, $t0, 1
	0x00400034	0x0401fff5	bgez $0,0xfffffff5	26:	b while
	0x00400038	0x2402000a	addiu $2,$0,0x00...	29:	li $v0, 10
	0x0040003c	0x0000000c	syscall	30:	syscall

Name	Number	Value
$zero	0	0x00000000
$at	1	0x10010000
$v0	2	0x00000001
$v1	3	0x00000000
$a0	4	0x0000001e
$a1	5	0x00000000
$a2	6	0x00000000
$a3	7	0x00000000
$t0	8	0x00000002
$t1	9	0x10010000
$t2	10	0x0000001e
$t3	11	0x10010002
$t4	12	0x00000000
$t5	13	0x00000000
$t6	14	0x00000000
$t7	15	0x00000000
$s0	16	0x00000000
$s1	17	0x00000000
$s2	18	0x00000000
$s3	19	0x00000000
$s4	20	0x00000000
$s5	21	0x00000000
$s6	22	0x00000000
$s7	23	0x00000000
$t8	24	0x00000000
$t9	25	0x00000000
$k0	26	0x00000000
$k1	27	0x00000000
$gp	28	0x10008000
$sp	29	0x7fffeffc
$fp	30	0x00000000
$ra	31	0x00000000
pc		0x0040001c
hi		0x00000000
lo		0x00000000

Imprime 30!

Passo 31

Text Segment

Bkpt	Address	Code	Basic		Source
	0x00400000	0x00004021	addu $8,$0,$0	8:	move $t0, $zero
	0x00400004	0x3c011001	lui $1,0x00001001	9:	la $t1, vetor
	0x00400008	0x34290000	ori $9,$1,0x0000...		
	0x0040000c	0x01285821	addu $11,$9,$8	11:	addu $t3, $t1, $t0
	0x00400010	0x816a0000	lb $10,0x0000000...	12:	lb $t2, 0($t3)
	0x00400014	0x24020001	addiu $2,$0,0x00...	15:	li $v0, 1
	0x00400018	0x000a2021	addu $4,$0,$10	16:	move $a0, $t2
	0x0040001c	0x0000000c	syscall	17:	syscall
	0x00400020	0x2402000b	addiu $2,$0,0x00...	19:	li $v0, 11
	0x00400024	0x24040020	addiu $4,$0,0x00...	20:	addiu $a0, $zero, 32
	0x00400028	0x0000000c	syscall	21:	syscall
	0x0040002c	0x11400002	beq $10,$0,0x000...	24:	beq $t2, $zero, fim
	0x00400030	0x25080001	addiu $8,$8,0x00...	25:	addiu $t0, $t0, 1
	0x00400034	0x0401fff5	bgez $0,0xfffffff5	26:	b while
	0x00400038	0x2402000a	addiu $2,$0,0x00...	29:	li $v0, 10
	0x0040003c	0x0000000c	syscall	30:	syscall

Name	Number	Value
$zero	0	0x00000000
$at	1	0x10010000
$v0	2	0x00000001
$v1	3	0x00000000
$a0	4	0x0000001e
$a1	5	0x00000000
$a2	6	0x00000000
$a3	7	0x00000000
$t0	8	0x00000002
$t1	9	0x10010000
$t2	10	0x0000001e
$t3	11	0x10010002
$t4	12	0x00000000
$t5	13	0x00000000
$t6	14	0x00000000
$t7	15	0x00000000
$s0	16	0x00000000
$s1	17	0x00000000
$s2	18	0x00000000
$s3	19	0x00000000
$s4	20	0x00000000
$s5	21	0x00000000
$s6	22	0x00000000
$s7	23	0x00000000
$t8	24	0x00000000
$t9	25	0x00000000
$k0	26	0x00000000
$k1	27	0x00000000
$gp	28	0x10008000
$sp	29	0x7fffeffc
$fp	30	0x00000000
$ra	31	0x00000000
pc		0x00400020
hi		0x00000000
lo		0x00000000

Passo 32

Text Segment

Bkpt	Address	Code	Basic		Source
	0x00400000	0x00004021	addu $8,$0,$0	8:	move $t0, $zero
	0x00400004	0x3c011001	lui $1,0x00001001	9:	la $t1, vetor
	0x00400008	0x34290000	ori $9,$1,0x0000...		
	0x0040000c	0x01285821	addu $11,$9,$8	11:	addu $t3, $t1, $t0
	0x00400010	0x816a0000	lb $10,0x00000000...	12:	lb $t2, 0($t3)
	0x00400014	0x24020001	addiu $2,$0,0x00...	15:	li $v0, 1
	0x00400018	0x000a2021	addu $4,$0,$10	16:	move $a0, $t2
	0x0040001c	0x0000000c	syscall	17:	syscall
	0x00400020	0x2402000b	addiu $2,$0,0x00...	19:	li $v0, 11
	0x00400024	0x24040020	addiu $4,$0,0x00...	20:	addiu $a0, $zero, 32
	0x00400028	0x0000000c	syscall	21:	syscall
	0x0040002c	0x11400002	beq $10,$0,0x000...	24:	beq $t2, $zero, fim
	0x00400030	0x25080001	addiu $8,$8,0x00...	25:	addiu $t0, $t0, 1
	0x00400034	0x0401fff5	bgez $0,0xfffffff5	26:	b while
	0x00400038	0x2402000a	addiu $2,$0,0x00...	29:	li $v0, 10
	0x0040003c	0x0000000c	syscall	30:	syscall

Name	Number	Value
$zero	0	0x00000000
$at	1	0x10010000
$v0	2	0x0000000b
$v1	3	0x00000000
$a0	4	0x0000001e
$a1	5	0x00000000
$a2	6	0x00000000
$a3	7	0x00000000
$t0	8	0x00000002
$t1	9	0x10010000
$t2	10	0x0000001e
$t3	11	0x10010002
$t4	12	0x00000000
$t5	13	0x00000000
$t6	14	0x00000000
$t7	15	0x00000000
$s0	16	0x00000000
$s1	17	0x00000000
$s2	18	0x00000000
$s3	19	0x00000000
$s4	20	0x00000000
$s5	21	0x00000000
$s6	22	0x00000000
$s7	23	0x00000000
$t8	24	0x00000000
$t9	25	0x00000000
$k0	26	0x00000000
$k1	27	0x00000000
$gp	28	0x10008000
$sp	29	0x7fffeffc
$fp	30	0x00000000
$ra	31	0x00000000
pc		0x00400024
hi		0x00000000
lo		0x00000000

Passo 33

Text Segment

Bkpt	Address	Code	Basic		Source
	0x00400000	0x00004021	addu $8,$0,$0	8:	move $t0, $zero
	0x00400004	0x3c011001	lui $1,0x00001001	9:	la $t1, vetor
	0x00400008	0x34290000	ori $9,$1,0x0000...		
	0x0040000c	0x01285821	addu $11,$9,$8	11:	addu $t3, $t1, $t0
	0x00400010	0x816a0000	lb $10,0x0000000...	12:	lb $t2, 0($t3)
	0x00400014	0x24020001	addiu $2,$0,0x00...	15:	li $v0, 1
	0x00400018	0x000a2021	addu $4,$0,$10	16:	move $a0, $t2
	0x0040001c	0x0000000c	syscall	17:	syscall
	0x00400020	0x2402000b	addiu $2,$0,0x00...	19:	li $v0, 11
	0x00400024	0x24040020	addiu $4,$0,0x00...	20:	addiu $a0, $zero, 32
	0x00400028	0x0000000c	syscall	21:	syscall
	0x0040002c	0x11400002	beq $10,$0,0x000...	24:	beq $t2, $zero, fim
	0x00400030	0x25080001	addiu $8,$8,0x00...	25:	addiu $t0, $t0, 1
	0x00400034	0x0401fff5	bgez $0,0xfffffff5	26:	b while
	0x00400038	0x2402000a	addiu $2,$0,0x00...	29:	li $v0, 10
	0x0040003c	0x0000000c	syscall	30:	syscall

Name	Number	Value
$zero	0	0x00000000
$at	1	0x10010000
$v0	2	0x0000000b
$v1	3	0x00000000
$a0	4	0x00000020
$a1	5	0x00000000
$a2	6	0x00000000
$a3	7	0x00000000
$t0	8	0x00000002
$t1	9	0x10010000
$t2	10	0x0000001e
$t3	11	0x10010002
$t4	12	0x00000000
$t5	13	0x00000000
$t6	14	0x00000000
$t7	15	0x00000000
$s0	16	0x00000000
$s1	17	0x00000000
$s2	18	0x00000000
$s3	19	0x00000000
$s4	20	0x00000000
$s5	21	0x00000000
$s6	22	0x00000000
$s7	23	0x00000000
$t8	24	0x00000000
$t9	25	0x00000000
$k0	26	0x00000000
$k1	27	0x00000000
$gp	28	0x10008000
$sp	29	0x7fffeffc
$fp	30	0x00000000
$ra	31	0x00000000
pc		0x00400028
hi		0x00000000
lo		0x00000000

Passo 34

Text Segment

Bkpt	Address	Code	Basic		Source
	0x00400000	0x00004021	addu $8,$0,$0	8:	move $t0, $zero
	0x00400004	0x3c011001	lui $1,0x00001001	9:	la $t1, vetor
	0x00400008	0x34290000	ori $9,$1,0x0000...		
	0x0040000c	0x01285821	addu $11,$9,$8	11:	addu $t3, $t1, $t0
	0x00400010	0x816a0000	lb $10,0x00000000...	12:	lb $t2, 0($t3)
	0x00400014	0x24020001	addiu $2,$0,0x00...	15:	li $v0, 1
	0x00400018	0x000a2021	addu $4,$0,$10	16:	move $a0, $t2
	0x0040001c	0x0000000c	syscall	17:	syscall
	0x00400020	0x2402000b	addiu $2,$0,0x00...	19:	li $v0, 11
	0x00400024	0x24040020	addiu $4,$0,0x00...	20:	addiu $a0, $zero, 32
	0x00400028	0x0000000c	syscall	21:	syscall
	0x0040002c	0x11400002	beq $10,$0,0x000...	24:	beq $t2, $zero, fim
	0x00400030	0x25080001	addiu $8,$8,0x00...	25:	addiu $t0, $t0, 1
	0x00400034	0x0401fff5	bgez $0,0xfffffff5	26:	b while
	0x00400038	0x2402000a	addiu $2,$0,0x00...	29:	li $v0, 10
	0x0040003c	0x0000000c	syscall	30:	syscall

Name	Number	Value
$zero	0	0x00000000
$at	1	0x10010000
$v0	2	0x0000000b
$v1	3	0x00000000
$a0	4	0x00000020
$a1	5	0x00000000
$a2	6	0x00000000
$a3	7	0x00000000
$t0	8	0x00000002
$t1	9	0x10010000
$t2	10	0x0000001e
$t3	11	0x10010002
$t4	12	0x00000000
$t5	13	0x00000000
$t6	14	0x00000000
$t7	15	0x00000000
$s0	16	0x00000000
$s1	17	0x00000000
$s2	18	0x00000000
$s3	19	0x00000000
$s4	20	0x00000000
$s5	21	0x00000000
$s6	22	0x00000000
$s7	23	0x00000000
$t8	24	0x00000000
$t9	25	0x00000000
$k0	26	0x00000000
$k1	27	0x00000000
$gp	28	0x10008000
$sp	29	0x7fffeffc
$fp	30	0x00000000
$ra	31	0x00000000
pc		0x0040002c
hi		0x00000000
lo		0x00000000

Passo 35

Text Segment

Bkpt	Address	Code	Basic		Source
	0x00400000	0x00004021	addu $8,$0,$0	8:	move $t0, $zero
	0x00400004	0x3c011001	lui $1,0x00001001	9:	la $t1, vetor
	0x00400008	0x34290000	ori $9,$1,0x0000...		
	0x0040000c	0x01285821	addu $11,$9,$8	11:	addu $t3, $t1, $t0
	0x00400010	0x816a0000	lb $10,0x0000000...	12:	lb $t2, 0($t3)
	0x00400014	0x24020001	addiu $2,$0,0x00...	15:	li $v0, 1
	0x00400018	0x000a2021	addu $4,$0,$10	16:	move $a0, $t2
	0x0040001c	0x0000000c	syscall	17:	syscall
	0x00400020	0x2402000b	addiu $2,$0,0x00...	19:	li $v0, 11
	0x00400024	0x24040020	addiu $4,$0,0x00...	20:	addiu $a0, $zero, 32
	0x00400028	0x0000000c	syscall	21:	syscall
	0x0040002c	0x11400002	beq $10,$0,0x000...	24:	beq $t2, $zero, fim
	0x00400030	0x25080001	addiu $8,$8,0x00...	25:	addiu $t0, $t0, 1
	0x00400034	0x0401fff5	bgez $0,0xfffffff5	26:	b while
	0x00400038	0x2402000a	addiu $2,$0,0x00...	29:	li $v0, 10
	0x0040003c	0x0000000c	syscall	30:	syscall

Name	Number	Value
$zero	0	0x00000000
$at	1	0x10010000
$v0	2	0x0000000b
$v1	3	0x00000000
$a0	4	0x00000020
$a1	5	0x00000000
$a2	6	0x00000000
$a3	7	0x00000000
$t0	8	0x00000002
$t1	9	0x10010000
$t2	10	0x0000001e
$t3	11	0x10010002
$t4	12	0x00000000
$t5	13	0x00000000
$t6	14	0x00000000
$t7	15	0x00000000
$s0	16	0x00000000
$s1	17	0x00000000
$s2	18	0x00000000
$s3	19	0x00000000
$s4	20	0x00000000
$s5	21	0x00000000
$s6	22	0x00000000
$s7	23	0x00000000
$t8	24	0x00000000
$t9	25	0x00000000
$k0	26	0x00000000
$k1	27	0x00000000
$gp	28	0x10008000
$sp	29	0x7fffeffc
$fp	30	0x00000000
$ra	31	0x00000000
pc		0x00400030
hi		0x00000000
lo		0x00000000

Passo 36

Text Segment

Bkpt	Address	Code	Basic		Source
	0x00400000	0x00004021	addu $8,$0,$0	8:	move $t0, $zero
	0x00400004	0x3c011001	lui $1,0x00001001	9:	la $t1, vetor
	0x00400008	0x34290000	ori $9,$1,0x0000...		
	0x0040000c	0x01285821	addu $11,$9,$8	11:	addu $t3, $t1, $t0
	0x00400010	0x816a0000	lb $10,0x0000000...	12:	lb $t2, 0($t3)
	0x00400014	0x24020001	addiu $2,$0,0x00...	15:	li $v0, 1
	0x00400018	0x000a2021	addu $4,$0,$10	16:	move $a0, $t2
	0x0040001c	0x0000000c	syscall	17:	syscall
	0x00400020	0x2402000b	addiu $2,$0,0x00...	19:	li $v0, 11
	0x00400024	0x24040020	addiu $4,$0,0x00...	20:	addiu $a0, $zero, 32
	0x00400028	0x0000000c	syscall	21:	syscall
	0x0040002c	0x11400002	beq $10,$0,0x000...	24:	beq $t2, $zero, fim
	0x00400030	0x25080001	addiu $8,$8,0x00...	25:	addiu $t0, $t0, 1
	0x00400034	0x0401fff5	bgez $0,0xfffffff5	26:	b while
	0x00400038	0x2402000a	addiu $2,$0,0x00...	29:	li $v0, 10
	0x0040003c	0x0000000c	syscall	30:	syscall

Name	Number	Value
$zero	0	0x00000000
$at	1	0x10010000
$v0	2	0x0000000b
$v1	3	0x00000000
$a0	4	0x00000020
$a1	5	0x00000000
$a2	6	0x00000000
$a3	7	0x00000000
$t0	8	0x00000003
$t1	9	0x10010000
$t2	10	0x0000001e
$t3	11	0x10010002
$t4	12	0x00000000
$t5	13	0x00000000
$t6	14	0x00000000
$t7	15	0x00000000
$s0	16	0x00000000
$s1	17	0x00000000
$s2	18	0x00000000
$s3	19	0x00000000
$s4	20	0x00000000
$s5	21	0x00000000
$s6	22	0x00000000
$s7	23	0x00000000
$t8	24	0x00000000
$t9	25	0x00000000
$k0	26	0x00000000
$k1	27	0x00000000
$gp	28	0x10008000
$sp	29	0x7fffeffc
$fp	30	0x00000000
$ra	31	0x00000000
pc		0x00400034
hi		0x00000000
lo		0x00000000

Passo 37

Text Segment

Bkpt	Address	Code	Basic		Source
	0x00400000	0x00004021	addu $8,$0,$0	8:	move $t0, $zero
	0x00400004	0x3c011001	lui $1,0x00001001	9:	la $t1, vetor
	0x00400008	0x34290000	ori $9,$1,0x0000...		
	0x0040000c	0x01285821	addu $11,$9,$8	11:	addu $t3, $t1, $t0
	0x00400010	0x816a0000	lb $10,0x0000000...	12:	lb $t2, 0($t3)
	0x00400014	0x24020001	addiu $2,$0,0x00...	15:	li $v0, 1
	0x00400018	0x000a2021	addu $4,$0,$10	16:	move $a0, $t2
	0x0040001c	0x0000000c	syscall	17:	syscall
	0x00400020	0x2402000b	addiu $2,$0,0x00...	19:	li $v0, 11
	0x00400024	0x24040020	addiu $4,$0,0x00...	20:	addiu $a0, $zero, 32
	0x00400028	0x0000000c	syscall	21:	syscall
	0x0040002c	0x11400002	beq $10,$0,0x000...	24:	beq $t2, $zero, fim
	0x00400030	0x25080001	addiu $8,$8,0x00...	25:	addiu $t0, $t0, 1
	0x00400034	0x0401fff5	bgez $0,0xfffffff5	26:	b while
	0x00400038	0x2402000a	addiu $2,$0,0x00...	29:	li $v0, 10
	0x0040003c	0x0000000c	syscall	30:	syscall

Name	Number	Value
$zero	0	0x00000000
$at	1	0x10010000
$v0	2	0x0000000b
$v1	3	0x00000000
$a0	4	0x00000020
$a1	5	0x00000000
$a2	6	0x00000000
$a3	7	0x00000000
$t0	8	0x00000003
$t1	9	0x10010000
$t2	10	0x0000001e
$t3	11	0x10010002
$t4	12	0x00000000
$t5	13	0x00000000
$t6	14	0x00000000
$t7	15	0x00000000
$s0	16	0x00000000
$s1	17	0x00000000
$s2	18	0x00000000
$s3	19	0x00000000
$s4	20	0x00000000
$s5	21	0x00000000
$s6	22	0x00000000
$s7	23	0x00000000
$t8	24	0x00000000
$t9	25	0x00000000
$k0	26	0x00000000
$k1	27	0x00000000
$gp	28	0x10008000
$sp	29	0x7fffeffc
$fp	30	0x00000000
$ra	31	0x00000000
pc		0x0040000c
hi		0x00000000
lo		0x00000000

Passo 38

Text Segment

Bkpt	Address	Code	Basic		Source
	0x00400000	0x00004021	addu $8,$0,$0	8:	move $t0, $zero
	0x00400004	0x3c011001	lui $1,0x00001001	9:	la $t1, vetor
	0x00400008	0x34290000	ori $9,$1,0x0000...		
	0x0040000c	0x01285821	addu $11,$9,$8	11:	addu $t3, $t1, $t0
	0x00400010	0x816a0000	lb $10,0x0000000...	12:	lb $t2, 0($t3)
	0x00400014	0x24020001	addiu $2,$0,0x00...	15:	li $v0, 1
	0x00400018	0x000a2021	addu $4,$0,$10	16:	move $a0, $t2
	0x0040001c	0x0000000c	syscall	17:	syscall
	0x00400020	0x2402000b	addiu $2,$0,0x00...	19:	li $v0, 11
	0x00400024	0x24040020	addiu $4,$0,0x00...	20:	addiu $a0, $zero, 32
	0x00400028	0x0000000c	syscall	21:	syscall
	0x0040002c	0x11400002	beq $10,$0,0x000...	24:	beq $t2, $zero, fim
	0x00400030	0x25080001	addiu $8,$8,0x00...	25:	addiu $t0, $t0, 1
	0x00400034	0x0401fff5	bgez $0,0xfffffff5	26:	b while
	0x00400038	0x2402000a	addiu $2,$0,0x00...	29:	li $v0, 10
	0x0040003c	0x0000000c	syscall	30:	syscall

Name	Number	Value
$zero	0	0x00000000
$at	1	0x10010000
$v0	2	0x0000000b
$v1	3	0x00000000
$a0	4	0x00000020
$a1	5	0x00000000
$a2	6	0x00000000
$a3	7	0x00000000
$t0	8	0x00000003
$t1	9	0x10010000
$t2	10	0x0000001e
$t3	11	0x10010003
$t4	12	0x00000000
$t5	13	0x00000000
$t6	14	0x00000000
$t7	15	0x00000000
$s0	16	0x00000000
$s1	17	0x00000000
$s2	18	0x00000000
$s3	19	0x00000000
$s4	20	0x00000000
$s5	21	0x00000000
$s6	22	0x00000000
$s7	23	0x00000000
$t8	24	0x00000000
$t9	25	0x00000000
$k0	26	0x00000000
$k1	27	0x00000000
$gp	28	0x10008000
$sp	29	0x7fffeffc
$fp	30	0x00000000
$ra	31	0x00000000
pc		0x00400010
hi		0x00000000
lo		0x00000000

Passo 39

Text Segment

Bkpt	Address	Code	Basic	Source
	0x00400000	0x00004021	addu $8,$0,$0	8: move $t0, $zero
	0x00400004	0x3c011001	lui $1,0x00001001	9: la $t1, vetor
	0x00400008	0x34290000	ori $9,$1,0x0000...	
	0x0040000c	0x01285821	addu $11,$9,$8	11: addu $t3, $t1, $t0
	0x00400010	0x816a0000	lb $10,0x0000000...	12: lb $t2, 0($t3)
	0x00400014	0x24020001	addiu $2,$0,0x00...	15: li $v0, 1
	0x00400018	0x000a2021	addu $4,$0,$10	16: move $a0, $t2
	0x0040001c	0x0000000c	syscall	17: syscall
	0x00400020	0x2402000b	addiu $2,$0,0x00...	19: li $v0, 11
	0x00400024	0x24040020	addiu $4,$0,0x00...	20: addiu $a0, $zero, 32
	0x00400028	0x0000000c	syscall	21: syscall
	0x0040002c	0x11400002	beq $10,$0,0x000...	24: beq $t2, $zero, fim
	0x00400030	0x25080001	addiu $8,$8,0x00...	25: addiu $t0, $t0, 1
	0x00400034	0x0401fff5	bgez $0,0xfffffff5	26: b while
	0x00400038	0x2402000a	addiu $2,$0,0x00...	29: li $v0, 10
	0x0040003c	0x0000000c	syscall	30: syscall

Name	Number	Value
$zero	0	0x00000000
$at	1	0x10010000
$v0	2	0x0000000b
$v1	3	0x00000000
$a0	4	0x00000020
$a1	5	0x00000000
$a2	6	0x00000000
$a3	7	0x00000000
$t0	8	0x00000003
$t1	9	0x10010000
$t2	10	0x00000000
$t3	11	0x10010003
$t4	12	0x00000000
$t5	13	0x00000000
$t6	14	0x00000000
$t7	15	0x00000000
$s0	16	0x00000000
$s1	17	0x00000000
$s2	18	0x00000000
$s3	19	0x00000000
$s4	20	0x00000000
$s5	21	0x00000000
$s6	22	0x00000000
$s7	23	0x00000000
$t8	24	0x00000000
$t9	25	0x00000000
$k0	26	0x00000000
$k1	27	0x00000000
$gp	28	0x10008000
$sp	29	0x7fffeffc
$fp	30	0x00000000
$ra	31	0x00000000
pc		0x00400014
hi		0x00000000
lo		0x00000000

Passo 40

Text Segment

Bkpt	Address	Code	Basic	Source
	0x00400000	0x00004021	addu $8,$0,$0	8: move $t0, $zero
	0x00400004	0x3c011001	lui $1,0x00001001	9: la $t1, vetor
	0x00400008	0x34290000	ori $9,$1,0x0000...	
	0x0040000c	0x01285821	addu $11,$9,$8	11: addu $t3, $t1, $t0
	0x00400010	0x816a0000	lb $10,0x0000000...	12: lb $t2, 0($t3)
	0x00400014	0x24020001	addiu $2,$0,0x00...	15: li $v0, 1
	0x00400018	0x000a2021	addu $4,$0,$10	16: move $a0, $t2
	0x0040001c	0x0000000c	syscall	17: syscall
	0x00400020	0x2402000b	addiu $2,$0,0x00...	19: li $v0, 11
	0x00400024	0x24040020	addiu $4,$0,0x00...	20: addiu $a0, $zero, 32
	0x00400028	0x0000000c	syscall	21: syscall
	0x0040002c	0x11400002	beq $10,$0,0x000...	24: beq $t2, $zero, fim
	0x00400030	0x25080001	addiu $8,$8,0x00...	25: addiu $t0, $t0, 1
	0x00400034	0x0401fff5	bgez $0,0xfffffff5	26: b while
	0x00400038	0x2402000a	addiu $2,$0,0x00...	29: li $v0, 10
	0x0040003c	0x0000000c	syscall	30: syscall

Name	Number	Value
$zero	0	0x00000000
$at	1	0x10010000
$v0	2	0x00000001
$v1	3	0x00000000
$a0	4	0x00000020
$a1	5	0x00000000
$a2	6	0x00000000
$a3	7	0x00000000
$t0	8	0x00000003
$t1	9	0x10010000
$t2	10	0x00000000
$t3	11	0x10010003
$t4	12	0x00000000
$t5	13	0x00000000
$t6	14	0x00000000
$t7	15	0x00000000
$s0	16	0x00000000
$s1	17	0x00000000
$s2	18	0x00000000
$s3	19	0x00000000
$s4	20	0x00000000
$s5	21	0x00000000
$s6	22	0x00000000
$s7	23	0x00000000
$t8	24	0x00000000
$t9	25	0x00000000
$k0	26	0x00000000
$k1	27	0x00000000
$gp	28	0x10008000
$sp	29	0x7fffeffc
$fp	30	0x00000000
$ra	31	0x00000000
pc		0x00400018
hi		0x00000000
lo		0x00000000

Passo 41

Text Segment

Bkpt	Address	Code	Basic		Source
	0x00400000	0x00004021	addu $8,$0,$0	8:	move $t0, $zero
	0x00400004	0x3c011001	lui $1,0x00001001	9:	la $t1, vetor
	0x00400008	0x34290000	ori $9,$1,0x0000...		
	0x0040000c	0x01285821	addu $11,$9,$8	11:	addu $t3, $t1, $t0
	0x00400010	0x816a0000	lb $10,0x0000000...	12:	lb $t2, 0($t3)
	0x00400014	0x24020001	addiu $2,$0,0x00...	15:	li $v0, 1
	0x00400018	0x000a2021	addu $4,$0,$10	16:	move $a0, $t2
	0x0040001c	0x0000000c	syscall	17:	syscall
	0x00400020	0x2402000b	addiu $2,$0,0x00...	19:	li $v0, 11
	0x00400024	0x24040020	addiu $4,$0,0x00...	20:	addiu $a0, $zero, 32
	0x00400028	0x0000000c	syscall	21:	syscall
	0x0040002c	0x11400002	beq $10,$0,0x000...	24:	beq $t2, $zero, fim
	0x00400030	0x25080001	addiu $8,$8,0x00...	25:	addiu $t0, $t0, 1
	0x00400034	0x0401fff5	bgez $0,0xfffffff5	26:	b while
	0x00400038	0x2402000a	addiu $2,$0,0x00...	29:	li $v0, 10
	0x0040003c	0x0000000c	syscall	30:	syscall

Name	Number	Value
$zero	0	0x00000000
$at	1	0x10010000
$v0	2	0x00000001
$v1	3	0x00000000
$a0	4	0x00000000
$a1	5	0x00000000
$a2	6	0x00000000
$a3	7	0x00000000
$t0	8	0x00000003
$t1	9	0x10010000
$t2	10	0x00000000
$t3	11	0x10010003
$t4	12	0x00000000
$t5	13	0x00000000
$t6	14	0x00000000
$t7	15	0x00000000
$s0	16	0x00000000
$s1	17	0x00000000
$s2	18	0x00000000
$s3	19	0x00000000
$s4	20	0x00000000
$s5	21	0x00000000
$s6	22	0x00000000
$s7	23	0x00000000
$t8	24	0x00000000
$t9	25	0x00000000
$k0	26	0x00000000
$k1	27	0x00000000
$gp	28	0x10008000
$sp	29	0x7fffeffc
$fp	30	0x00000000
$ra	31	0x00000000
pc		0x0040001c
hi		0x00000000
lo		0x00000000

Imprime 0!

Passo 42

Text Segment

Bkpt	Address	Code	Basic		Source
	0x00400000	0x00004021	addu $8,$0,$0	8:	move $t0, $zero
	0x00400004	0x3c011001	lui $1,0x00001001	9:	la $t1, vetor
	0x00400008	0x34290000	ori $9,$1,0x0000...		
	0x0040000c	0x01285821	addu $11,$9,$8	11:	addu $t3, $t1, $t0
	0x00400010	0x816a0000	lb $10,0x0000000...	12:	lb $t2, 0($t3)
	0x00400014	0x24020001	addiu $2,$0,0x00...	15:	li $v0, 1
	0x00400018	0x000a2021	addu $4,$0,$10	16:	move $a0, $t2
	0x0040001c	0x0000000c	syscall	17:	syscall
	0x00400020	0x2402000b	addiu $2,$0,0x00...	19:	li $v0, 11
	0x00400024	0x24040020	addiu $4,$0,0x00...	20:	addiu $a0, $zero, 32
	0x00400028	0x0000000c	syscall	21:	syscall
	0x0040002c	0x11400002	beq $10,$0,0x000...	24:	beq $t2, $zero, fim
	0x00400030	0x25080001	addiu $8,$8,0x00...	25:	addiu $t0, $t0, 1
	0x00400034	0x0401fff5	bgez $0,0xfffffff5	26:	b while
	0x00400038	0x2402000a	addiu $2,$0,0x00...	29:	li $v0, 10
	0x0040003c	0x0000000c	syscall	30:	syscall

Name	Number	Value
$zero	0	0x00000000
$at	1	0x10010000
$v0	2	0x00000001
$v1	3	0x00000000
$a0	4	0x00000000
$a1	5	0x00000000
$a2	6	0x00000000
$a3	7	0x00000000
$t0	8	0x00000003
$t1	9	0x10010000
$t2	10	0x00000000
$t3	11	0x10010003
$t4	12	0x00000000
$t5	13	0x00000000
$t6	14	0x00000000
$t7	15	0x00000000
$s0	16	0x00000000
$s1	17	0x00000000
$s2	18	0x00000000
$s3	19	0x00000000
$s4	20	0x00000000
$s5	21	0x00000000
$s6	22	0x00000000
$s7	23	0x00000000
$t8	24	0x00000000
$t9	25	0x00000000
$k0	26	0x00000000
$k1	27	0x00000000
$gp	28	0x10008000
$sp	29	0x7fffeffc
$fp	30	0x00000000
$ra	31	0x00000000
pc		0x00400020
hi		0x00000000
lo		0x00000000

Passo 43

Text Segment

Bkpt	Address	Code	Basic		Source
	0x00400000	0x00004021	addu $8,$0,$0	8:	move $t0, $zero
	0x00400004	0x3c011001	lui $1,0x00001001	9:	la $t1, vetor
	0x00400008	0x34290000	ori $9,$1,0x0000...		
	0x0040000c	0x01285821	addu $11,$9,$8	11:	addu $t3, $t1, $t0
	0x00400010	0x816a0000	lb $10,0x0000000...	12:	lb $t2, 0($t3)
	0x00400014	0x24020001	addiu $2,$0,0x00...	15:	li $v0, 1
	0x00400018	0x000a2021	addu $4,$0,$10	16:	move $a0, $t2
	0x0040001c	0x0000000c	syscall	17:	syscall
	0x00400020	0x2402000b	addiu $2,$0,0x00...	19:	li $v0, 11
	0x00400024	0x24040020	addiu $4,$0,0x00...	20:	addiu $a0, $zero, 32
	0x00400028	0x0000000c	syscall	21:	syscall
	0x0040002c	0x11400002	beq $10,$0,0x000...	24:	beq $t2, $zero, fim
	0x00400030	0x25080001	addiu $8,$8,0x00...	25:	addiu $t0, $t0, 1
	0x00400034	0x0401fff5	bgez $0,0xfffffff5	26:	b while
	0x00400038	0x2402000a	addiu $2,$0,0x00...	29:	li $v0, 10
	0x0040003c	0x0000000c	syscall	30:	syscall

Name	Number	Value
$zero	0	0x00000000
$at	1	0x10010000
$v0	2	0x0000000b
$v1	3	0x00000000
$a0	4	0x00000000
$a1	5	0x00000000
$a2	6	0x00000000
$a3	7	0x00000000
$t0	8	0x00000003
$t1	9	0x10010000
$t2	10	0x00000000
$t3	11	0x10010003
$t4	12	0x00000000
$t5	13	0x00000000
$t6	14	0x00000000
$t7	15	0x00000000
$s0	16	0x00000000
$s1	17	0x00000000
$s2	18	0x00000000
$s3	19	0x00000000
$s4	20	0x00000000
$s5	21	0x00000000
$s6	22	0x00000000
$s7	23	0x00000000
$t8	24	0x00000000
$t9	25	0x00000000
$k0	26	0x00000000
$k1	27	0x00000000
$gp	28	0x10008000
$sp	29	0x7fffeffc
$fp	30	0x00000000
$ra	31	0x00000000
pc		0x00400024
hi		0x00000000
lo		0x00000000

Passo 44

Text Segment

Bkpt	Address	Code	Basic		Source
	0x00400000	0x00004021	addu $8,$0,$0	8:	move $t0, $zero
	0x00400004	0x3c011001	lui $1,0x00001001	9:	la $t1, vetor
	0x00400008	0x34290000	ori $9,$1,0x0000...		
	0x0040000c	0x01285821	addu $11,$9,$8	11:	addu $t3, $t1, $t0
	0x00400010	0x816a0000	lb $10,0x0000000...	12:	lb $t2, 0($t3)
	0x00400014	0x24020001	addiu $2,$0,0x00...	15:	li $v0, 1
	0x00400018	0x000a2021	addu $4,$0,$10	16:	move $a0, $t2
	0x0040001c	0x0000000c	syscall	17:	syscall
	0x00400020	0x2402000b	addiu $2,$0,0x00...	19:	li $v0, 11
	0x00400024	0x24040020	addiu $4,$0,0x00...	20:	addiu $a0, $zero, 32
	0x00400028	0x0000000c	syscall	21:	syscall
	0x0040002c	0x11400002	beq $10,$0,0x000...	24:	beq $t2, $zero, fim
	0x00400030	0x25080001	addiu $8,$8,0x00...	25:	addiu $t0, $t0, 1
	0x00400034	0x0401fff5	bgez $0,0xfffffff5	26:	b while
	0x00400038	0x2402000a	addiu $2,$0,0x00...	29:	li $v0, 10
	0x0040003c	0x0000000c	syscall	30:	syscall

Name	Number	Value
$zero	0	0x00000000
$at	1	0x10010000
$v0	2	0x0000000b
$v1	3	0x00000000
$a0	4	0x00000020
$a1	5	0x00000000
$a2	6	0x00000000
$a3	7	0x00000000
$t0	8	0x00000003
$t1	9	0x10010000
$t2	10	0x00000000
$t3	11	0x10010003
$t4	12	0x00000000
$t5	13	0x00000000
$t6	14	0x00000000
$t7	15	0x00000000
$s0	16	0x00000000
$s1	17	0x00000000
$s2	18	0x00000000
$s3	19	0x00000000
$s4	20	0x00000000
$s5	21	0x00000000
$s6	22	0x00000000
$s7	23	0x00000000
$t8	24	0x00000000
$t9	25	0x00000000
$k0	26	0x00000000
$k1	27	0x00000000
$gp	28	0x10008000
$sp	29	0x7fffeffc
$fp	30	0x00000000
$ra	31	0x00000000
pc		0x00400028
hi		0x00000000
lo		0x00000000

Passo 45

Text Segment

Bkpt	Address	Code	Basic		Source
	0x00400000	0x00004021	addu $8,$0,$0	8:	move $t0, $zero
	0x00400004	0x3c011001	lui $1,0x00001001	9:	la $t1, vetor
	0x00400008	0x34290000	ori $9,$1,0x0000...		
	0x0040000c	0x01285821	addu $11,$9,$8	11:	addu $t3, $t1, $t0
	0x00400010	0x816a0000	lb $10,0x0000000...	12:	lb $t2, 0($t3)
	0x00400014	0x24020001	addiu $2,$0,0x00...	15:	li $v0, 1
	0x00400018	0x000a2021	addu $4,$0,$10	16:	move $a0, $t2
	0x0040001c	0x0000000c	syscall	17:	syscall
	0x00400020	0x2402000b	addiu $2,$0,0x00...	19:	li $v0, 11
	0x00400024	0x24040020	addiu $4,$0,0x00...	20:	addiu $a0, $zero, 32
	0x00400028	0x0000000c	syscall	21:	syscall
	0x0040002c	0x11400002	beq $10,$0,0x000...	24:	beq $t2, $zero, fim
	0x00400030	0x25080001	addiu $8,$8,0x00...	25:	addiu $t0, $t0, 1
	0x00400034	0x0401fff5	bgez $0,0xfffffff5	26:	b while
	0x00400038	0x2402000a	addiu $2,$0,0x00...	29:	li $v0, 10
	0x0040003c	0x0000000c	syscall	30:	syscall

Name	Number	Value
$zero	0	0x00000000
$at	1	0x10010000
$v0	2	0x0000000b
$v1	3	0x00000000
$a0	4	0x00000020
$a1	5	0x00000000
$a2	6	0x00000000
$a3	7	0x00000000
$t0	8	0x00000003
$t1	9	0x10010000
$t2	10	0x00000000
$t3	11	0x10010003
$t4	12	0x00000000
$t5	13	0x00000000
$t6	14	0x00000000
$t7	15	0x00000000
$s0	16	0x00000000
$s1	17	0x00000000
$s2	18	0x00000000
$s3	19	0x00000000
$s4	20	0x00000000
$s5	21	0x00000000
$s6	22	0x00000000
$s7	23	0x00000000
$t8	24	0x00000000
$t9	25	0x00000000
$k0	26	0x00000000
$k1	27	0x00000000
$gp	28	0x10008000
$sp	29	0x7fffeffc
$fp	30	0x00000000
$ra	31	0x00000000
pc		0x0040002c
hi		0x00000000
lo		0x00000000

Passo 46

Text Segment

Bkpt	Address	Code	Basic		Source
	0x00400000	0x00004021	addu $8,$0,$0	8:	move $t0, $zero
	0x00400004	0x3c011001	lui $1,0x00001001	9:	la $t1, vetor
	0x00400008	0x34290000	ori $9,$1,0x0000...		
	0x0040000c	0x01285821	addu $11,$9,$8	11:	addu $t3, $t1, $t0
	0x00400010	0x816a0000	lb $10,0x0000000...	12:	lb $t2, 0($t3)
	0x00400014	0x24020001	addiu $2,$0,0x00...	15:	li $v0, 1
	0x00400018	0x000a2021	addu $4,$0,$10	16:	move $a0, $t2
	0x0040001c	0x0000000c	syscall	17:	syscall
	0x00400020	0x2402000b	addiu $2,$0,0x00...	19:	li $v0, 11
	0x00400024	0x24040020	addiu $4,$0,0x00...	20:	addiu $a0, $zero, 32
	0x00400028	0x0000000c	syscall	21:	syscall
	0x0040002c	0x11400002	beq $10,$0,0x000...	24:	beq $t2, $zero, fim
	0x00400030	0x25080001	addiu $8,$8,0x00...	25:	addiu $t0, $t0, 1
	0x00400034	0x0401fff5	bgez $0,0xfffffff5	26:	b while
	0x00400038	0x2402000a	addiu $2,$0,0x00...	29:	li $v0, 10
	0x0040003c	0x0000000c	syscall	30:	syscall

Name	Number	Value
$zero	0	0x00000000
$at	1	0x10010000
$v0	2	0x0000000b
$v1	3	0x00000000
$a0	4	0x00000020
$a1	5	0x00000000
$a2	6	0x00000000
$a3	7	0x00000000
$t0	8	0x00000003
$t1	9	0x10010000
$t2	10	0x00000000
$t3	11	0x10010003
$t4	12	0x00000000
$t5	13	0x00000000
$t6	14	0x00000000
$t7	15	0x00000000
$s0	16	0x00000000
$s1	17	0x00000000
$s2	18	0x00000000
$s3	19	0x00000000
$s4	20	0x00000000
$s5	21	0x00000000
$s6	22	0x00000000
$s7	23	0x00000000
$t8	24	0x00000000
$t9	25	0x00000000
$k0	26	0x00000000
$k1	27	0x00000000
$gp	28	0x10008000
$sp	29	0x7fffeffc
$fp	30	0x00000000
$ra	31	0x00000000
pc		0x00400038
hi		0x00000000
lo		0x00000000

Passo 47

Text Segment

Bkpt	Address	Code	Basic		Source
	0x00400000	0x00004021	addu $8,$0,$0	8:	move $t0, $zero
	0x00400004	0x3c011001	lui $1,0x00001001	9:	la $t1, vetor
	0x00400008	0x34290000	ori $9,$1,0x0000...		
	0x0040000c	0x01285821	addu $11,$9,$8	11:	addu $t3, $t1, $t0
	0x00400010	0x816a0000	lb $10,0x00000000...	12:	lb $t2, 0($t3)
	0x00400014	0x24020001	addiu $2,$0,0x00...	15:	li $v0, 1
	0x00400018	0x000a2021	addu $4,$0,$10	16:	move $a0, $t2
	0x0040001c	0x0000000c	syscall	17:	syscall
	0x00400020	0x2402000b	addiu $2,$0,0x00...	19:	li $v0, 11
	0x00400024	0x24040020	addiu $4,$0,0x00...	20:	addiu $a0, $zero, 32
	0x00400028	0x0000000c	syscall	21:	syscall
	0x0040002c	0x11400002	beq $10,$0,0x000...	24:	beq $t2, $zero, fim
	0x00400030	0x25080001	addiu $8,$8,0x00...	25:	addiu $t0, $t0, 1
	0x00400034	0x0401fff5	bgez $0,0xfffffff5	26:	b while
	0x00400038	0x2402000a	addiu $2,$0,0x00...	29:	li $v0, 10
	0x0040003c	0x0000000c	syscall	30:	syscall

Name	Number	Value
$zero	0	0x00000000
$at	1	0x10010000
$v0	2	0x0000000a
$v1	3	0x00000000
$a0	4	0x00000020
$a1	5	0x00000000
$a2	6	0x00000000
$a3	7	0x00000000
$t0	8	0x00000003
$t1	9	0x10010000
$t2	10	0x00000000
$t3	11	0x10010003
$t4	12	0x00000000
$t5	13	0x00000000
$t6	14	0x00000000
$t7	15	0x00000000
$s0	16	0x00000000
$s1	17	0x00000000
$s2	18	0x00000000
$s3	19	0x00000000
$s4	20	0x00000000
$s5	21	0x00000000
$s6	22	0x00000000
$s7	23	0x00000000
$t8	24	0x00000000
$t9	25	0x00000000
$k0	26	0x00000000
$k1	27	0x00000000
$gp	28	0x10008000
$sp	29	0x7fffeffc
$fp	30	0x00000000
$ra	31	0x00000000
pc		0x0040003c
hi		0x00000000
lo		0x00000000

Saída do Programa

10 20 30 00

-- program is finished running -

Ponteiros em C

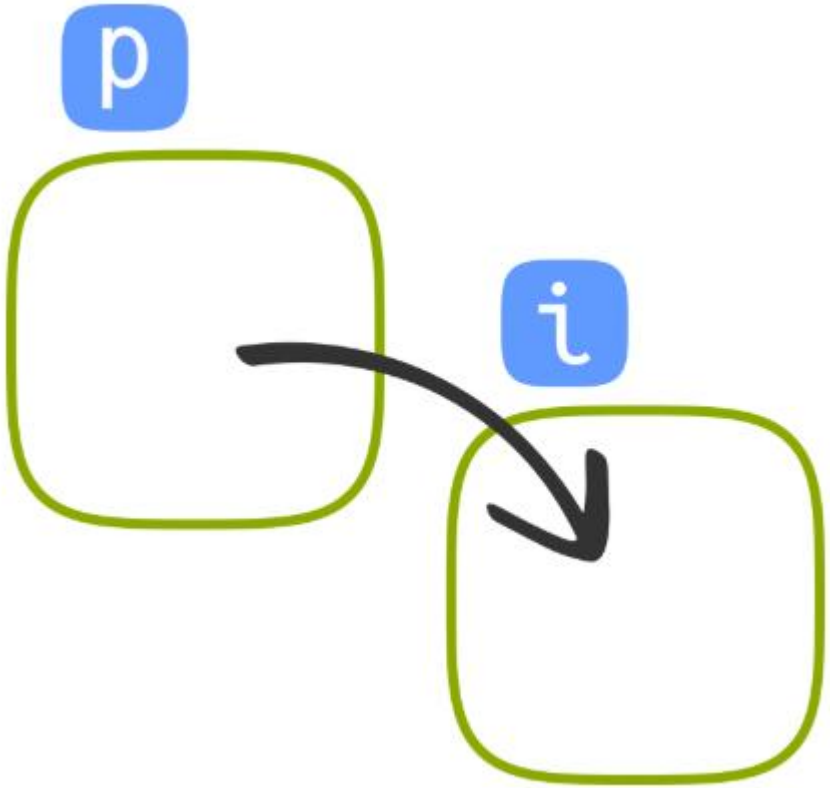
Declaração de ponteiros em C

Em C, um **ponteiro** é uma variável que armazena o **endereço de memória** de outra variável.

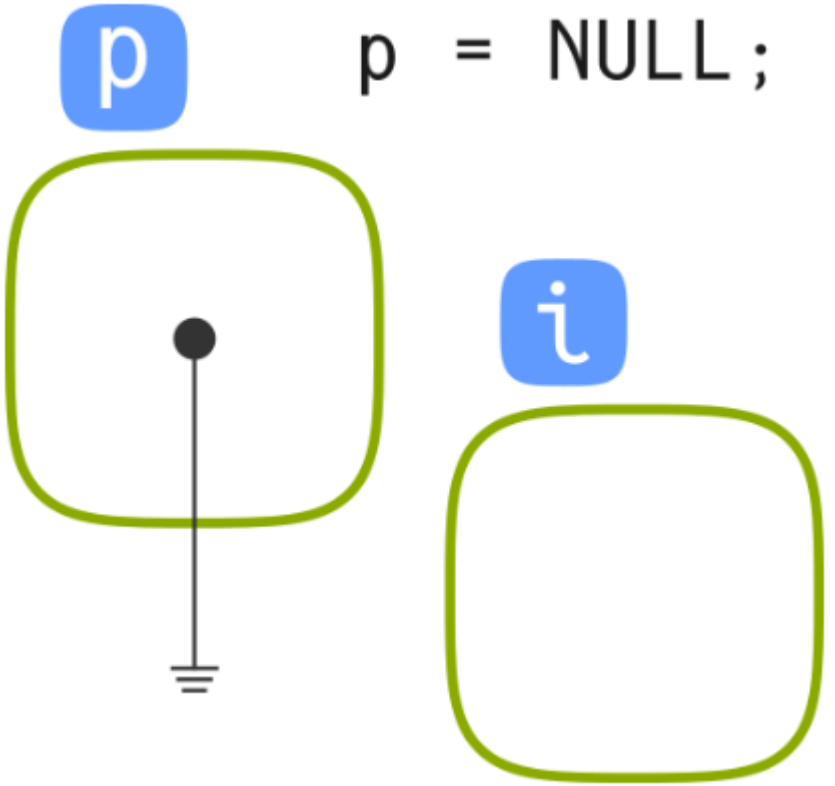
```
int *a;          // declara um ponteiro de inteiro “a”.
*a = 10;         // ao acessar o endereço de memória “a”, passa a
                 // existir o valor “10”
Int *b = a;      // “b” recebe o endereço de memória de “a”.

A = NULL;        // macro que é igual a 0, usada para iniciar ponteiros.
```

p = &i;



p = NULL;



Operador sizeof

```
#include <stdio.h>
int main(void) {
   printf("This computing environment uses:\n");
   printf("%zu byte for chars\n", sizeof(char));
   printf("%zu bytes for shorts\n", sizeof(short int));
   printf("%zu bytes for ints\n", sizeof(int));
   printf("%zu bytes for longs\n", sizeof(long int));
   printf("%zu bytes for long longs\n", sizeof(long long int));
   printf("%zu bytes for floats\n", sizeof(float));
   printf("%zu bytes for doubles\n", sizeof(double));
   printf("%zu bytes for pointers\n", sizeof(int *));
   return 0;
}
```

// %zu serve como especificador de formato na linguagem de programação C, projetado especificamente para imprimir variáveis do tipo size_t.

// size_t é um tipo inteiro sem sinal usado para representar o tamanho de objetos em bytes. Ele é comumente retornado pelo operador sizeof em C ao determinar o tamanho de um tipo de dado ou de uma variável.

Tabela de prioridades em C

1	`!`, `~` (type), `++`, `--`, `+`, `-`, `*`, `&`, `sizeof`
2	`*`, `/`, `%`
3	`+`, `-`
4	`<<`, `>>`
5	`<`, `<=`, `>`, `>=`
6	`==`, `!=`
7	`&`
8	`|`
9	`&&`
10	`||`
11	`=`, `+=`, `-=`, `*=`, `/=`, `%=`, `&=`, `^=`, `|=`, `>>=`, `<<=`

Exemplo 1: Declaração de ponteiros

```
#include <stdio.h>

int main() {
    int idade = 30;
    int *ptr = &idade;

    printf("Valor de idade: %d\n", idade);
    printf("Endereco de idade: %p\n", &idade);
    printf("Valor armazenado em ptr: %p\n", ptr);
    printf("Valor apontado por ptr: %d\n", *ptr);

    return 0;
}
```

Inicialização de ponteiros

Errado (ptr não aponta para uma área válida de memória)

```
int *ptr;
*ptr = 10;
```

Correto (Agora ptr aponta para x)

```
int x;
int *ptr = &x;
```

Exemplo 2: Alterando uma variável usando ponteiro

```
#include <stdio.h>

int main() {
    int numero = 10;
    int *ptr = &numero;

    printf("Antes: %d\n", numero);

    *ptr = 50;
    printf("Depois: %d\n", numero);

    return 0;
}
```

Passagem de variáveis por referência

Em C, é possível passar argumentos por valor (cópia) ou por referência (endereços de memória) através do uso de ponteiros. Ponteiros são passados para as funções como qualquer outra variável (porém, com sintaxe própria).

Exemplo:

```
void swap(int *x, int *y) {
        int temp;
        temp = *x;                  // salva o valor de x
        *x = *y;                    // faz x receber o valor de y
        *y = temp;                  // faz y receber o valor anterior de x
}
void main() {
        int i=10, j=20;
        swap(&i, &j);
}
```

Exemplo 3: Declaração de ponteiros e passagem de variáveis por referência

```
#include <stdio.h>

void play(int *nro) {
    *nro += 1;
    printf("play: %d\n", *nro);
}

int main() {
    int nro = 10;
    printf("main: %d\n", nro);
    nro += 1;
    printf("main: %d\n", nro);
    nro += 1;
    printf("main: %d\n", nro);
    nro += 1;
    play(&nro);
    printf("main: %d\n", nro);
    return 0;
}
```

Exemplo 4: Passagem de variáveis por valor VS por referência.

```
#include <stdio.h>

void play1(char a) {
    printf("Valor: <%c> Memoria <%d>\n", a, &a);
}

void play2(char *a) {
    printf("Valor: <%c> Memoria <%d>\n", *a, a);
}

int main() {
    char a = 'A';
    printf("Valor: <%c> Memoria <%d>\n", a, &a);
    play1(a);
    play2(&a);

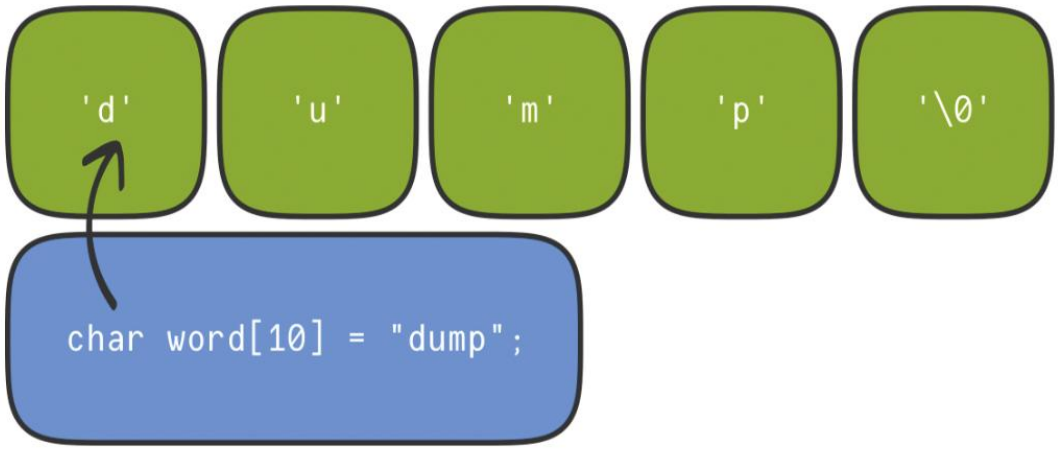
    return 0;
}
```

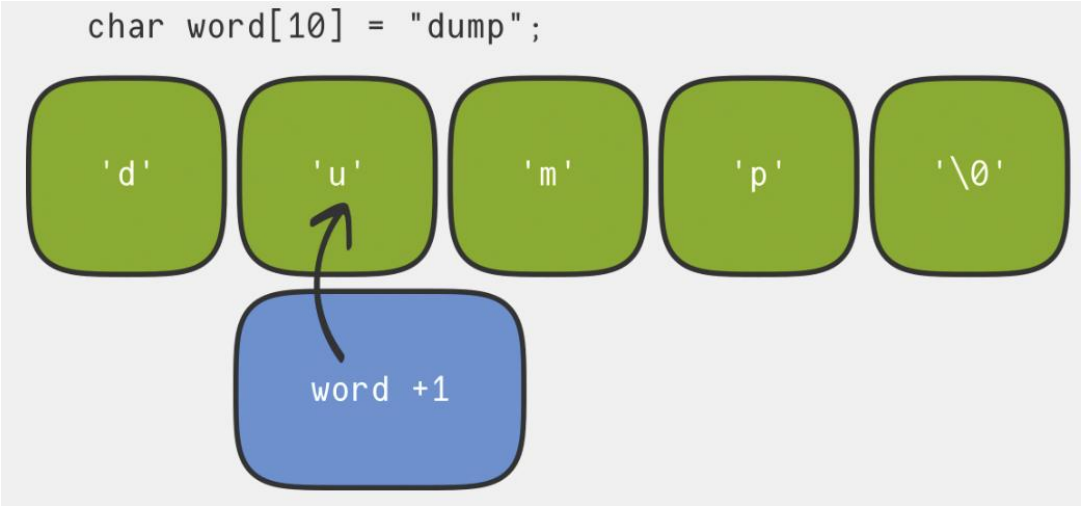
Exemplo 5: Passagem de arrays por referência

```
#include <stdio.h>
void dobrarElementos(int v[], int tamanho) {
        for (int i = 0; i < tamanho; i++) {
                v[i] = v[i] * 2;
        }
}
int main() {
        int numeros[5] = {1, 2, 3, 4, 5};
        dobrarElementos(numeros, 5);
        printf("Array após a função:\n");
        for (int i = 0; i < 5; i++) {
                printf("%d ", numeros[i]);
        }
        return 0;
}
```

Indexação de Arrays

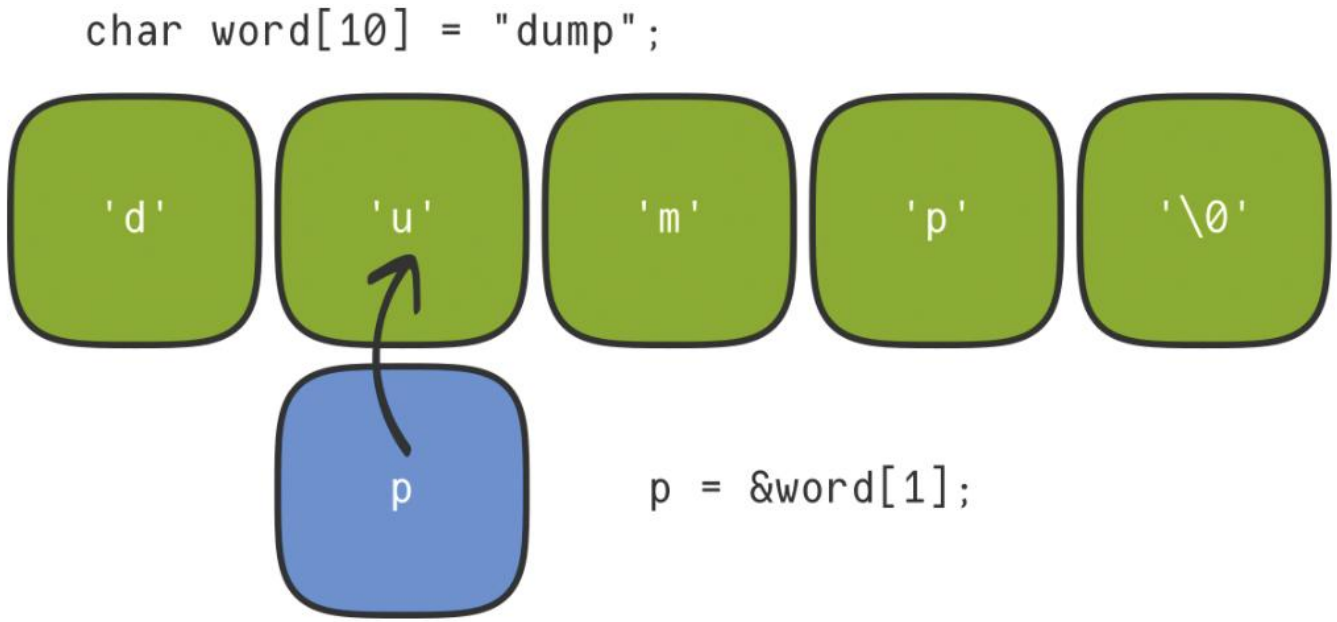
*(word + 1) = 'a';
igual a
word[1] = 'a';





Indexação de Arrays

```
char string[] = "ABC";
char *p;
```



Ponteiros em MIPS

Um ponteiro é uma variável que guarda um endereço de memória.

Em MIPS, um endereço possui 32 bits.

C: Ponteiro Global x Local

```
1  int *p1; // global
2
3  void main()
4  {
5     int *p2; // local
6     ...
7  }
```

MIPS: Declaração de Ponteiros Globais

Em C:

```
int *p1, *p2;          // Endereço de memória
char *p3;              // Endereço de memória
```

Em MIPS:

```
.data
p1: .word 0
p2: .word 0
p3: .word 0
```

OBS 1: Em MIPS, um endereço de memória deve sempre ter 32 bits (4 bytes).

OBS 2: Para Ponteiros Locais, utilizam-se os registradores.

MIPS: Inicialização de Ponteiros

```
1  int v[100];
2  char a='E', b='K';
3  char *pglob = &a;
```

```
1  v:      .space 400
2  a:      .byte 'E'
3  b:      .byte 'K'
4  pglob:.word a    # char *pglob = &a
```



Não funciona no MARS. Ver exemplo 2.

MIPS: Inicialização de Ponteiros

```
1  void f()
2  {
3     pglob = &b;
4  }
```

```
1  f:
2     la $t1, b    # &b
3     la $t2, pglob # &pglob
4     sw $t1, 0($t2)# pglob = &b
5     ...
```

MIPS: Inicialização de Ponteiros

```
1  void f()
2  {
3     int *ploc; /*      $t0 */
4     ploc = &b;
5  }
```

```
1  f:
2     la $t0, b   # ploc = &b
3     ...
```

MIPS: Dereference Operator

```
1  char *pglob;
2
3  void g()
4  {
5     char tmp; // a $t0
6     tmp = *pglob;
7     ...
8  }
```

```
1  g: ...
2     la $t2, pglob   # &pglob
3     lw $t3, 0($t2)  # pglob
4     lb $t0, 0($t3)  # *pglob
5     ...
```

MIPS: Dereference Operator

(writing)

```
1  char *pglob;
2
3  void g()
4  {
5     *pglob = 3;
6     ...
7  }
```

```
1  g:  ...
2     la  $t2, pglob   # &pglob
3     lw  $t3, 0($t2)  # pglob
4     li  $t4, 3
5     sb  $t4, 0($t3)  # *pglob = 3
6     ...
```

MIPS: Dereference Operator

```
1  void g()
2  {
3     char tmp;     //$t0
4     char *ploc;   //$t1
5     ...
6     tmp = *ploc;
7     ...
8  }
```

```
1  g:
2     ...
3     lb  $t0, 0($t1)
4     ...
```

MIPS: Dereference Operator

(writing)

```
1  void g()
2  {
3     char *ploc; //$t1
4     ...
5     *ploc = 3;
6     ...
7  }
```

```
1  g:
2     ...
3     li $t0, 3
4     sb $t0, 0($t1)
5     ...
```

MIPS: Aritmética de Ponteiros

```
1  char *p1;          //$t1
2  int *p2;           //$t2
3  long long *p3;     //$t3
4  ...
5  p1 = p1 + 3;
6  p2 = p2 + 3;
7  p3 = p3 + 3;
```

```
1  addiu $t1, $t1, 3
2  addiu $t2, $t2, 12
3  addiu $t3, $t3, 24
```

C: Relação entre Ponteiros e Vetores

```
1   int vec[100];
2   int *p;
3
4   main()
5   {
6       ...
7       p = vec;
8       p[8] = 10; /*  Element 8 of the vector          */
9       ...
10  }
```

Exemplo 2 - MARS

O exemplo a seguir mostra a execução de código MIPS no simulador MARS.

Exemplo 2

```
.data
        a: .byte 'A'
        b: .byte 'B'
        c: .byte 'C'
        d: .byte 'D'
        e: .byte 'E'
        f: .byte 'F'
        p1: .word a     # Prof: inicializar com 0.
.text
.globl main
main:
        # Carrega endereços em memória.
        la $t0, a
        la $t1, b
        la $t2, c
        la $t3, d
        la $t4, e
        la $t5, f
rep:    # Imprime: ABCDEF
        la $v0, 11
        lb $a0, 0($t0)
        syscall
        lb $a0, 0($t1)
        syscall
        lb $a0, 0($t2)
        syscall
        lb $a0, 0($t3)
        syscall
        lb $a0, 0($t4)
        syscall
        lb $a0, 0($t5)
        syscall
        # Substituindo 'A' por 'Z'
        la $t6, p1
        lw $t7, 0($t6)      # Recupera o endereço de a
        li $t8, 90          # Código ASCII de Z
        sb $t8, 0($t7)      # Modifica o caractere apontado
        lb $a0, 0($t7)
        li $v0, 11
        syscall
```

Saída do Programa

ABCDEFZ

-- program is finished running --

Exemplo 3 - MARS

O exemplo a seguir mostra a execução de código MIPS no simulador MARS.

Exemplo 3

```
.data
    numero: .word 10        # Variável inteira com valor 10
    ponteiro: .word 0       # Espaço para armazenar um endereço

.text
.globl main

main:
    # 1. Carrega o endereço da variável numero
    la $t0, numero

    # 2. Armazena esse endereço dentro de ponteiro
    la $t1, ponteiro
    sw $t0, 0($t1)

    # 3. Recupera o endereço armazenado em ponteiro
    lw $t2, 0($t1)

    # 4. Lê o valor da variável apontada
    lw $t3, 0($t2)          # $t3 = 10

    # 5. Incrementa o valor em 5
    addi $t3, $t3, 5        # $t3 = 15

    # 6. Escreve o novo valor na memória usando o ponteiro
    sw $t3, 0($t2)          # numero = 15

    # 7. Imprime o valor atualizado
    lw $a0, 0($t2)
    li $v0, 1               # Serviço 1: imprimir inteiro
    syscall                 # Imprime 15

    # 8. Encerra o programa
    li $v0, 10
    syscall
```

Saída do Programa

15

-- program is finished running --

Exemplo 4 - MARS

O exemplo a seguir mostra a execução de código MIPS no simulador MARS.

Exenplo 4

```
1  .data
2      numero: .word 10        # Variável que começa com 10
3
4  .text
5  .globl main
6
7  main:
8      # 1. Carrega o endereço de numero
9      la $t0, numero
10
11     # 2. Armazena o valor 50 na variável apontada
12     li $t1, 50
13     sw $t1, 0($t0)
14
15     # 3. Lê o novo valor usando o ponteiro
16     lw $a0, 0($t0)
17
18     # 4. Imprime o valor
19     li $v0, 1
20     syscall
21
22     # 5. Encerra o programa
23     li $v0, 10
24     syscall
```

Saída do Programa

50

-- program is finished running --

Atividade Avaliativa 5

Utilizando o Assembler NASM (Windows)

Exercício 1

Considere a declaração:

```
.data
vetor: .word 12, 25, 8, 17, 30, 14, 9, 21
```

Escreva um programa em Assembly MIPS que percorra o vetor e calcule a soma de todos os seus elementos.

Requisitos:

- Utilizar um laço de repetição.
- Utilizar registradores para o índice e o acumulador.
- Considerar que cada elemento ocupa 4 bytes.
- Imprimir a soma calculada.
- Resultado esperado: 136.

Exercício 2

Considere o vetor:

```
.data
vetor: .word 10, -5, 8, -12, 0, -3, 15, -1, 7, -9
```

Implemente um programa que conte quantos elementos negativos existem no vetor.

Requisitos:

- Percorrer todos os elementos.
- Utilizar comparações condicionais.
- Incrementar um contador somente quando o elemento for negativo.
- Imprimir a quantidade encontrada.

Exercício 3

Considere o vetor:

```
.data
vetor: .word 10, 20, 30, 40, 50, 60
```

Escreva um programa que inverta a ordem dos elementos do vetor na própria memória, sem declarar um segundo vetor.

O programa deverá produzir:

60 50 40 30 20 10

Exercício 4

Considere o vetor:

```
.data
vetor: .word 15, 8, 42, 19, 7, 42, 31, 12
```

Solicite ao usuário um número inteiro e procure sua ocorrência no vetor.

O programa deverá imprimir o índice da primeira ocorrência ou -1, caso o elemento não seja encontrado.

Exemplo: se o usuário informar 42, o programa deverá imprimir 2, considerando índices iniciados em zero.

Exercício 5

Considere o vetor:

```
.data
vetor: .word 35, 12, 48, 7, 23, 19
```

Implemente o algoritmo de ordenação por seleção (Selection Sort) em ordem crescente.

Requisitos:

- Encontrar o menor elemento da parte ainda não ordenada.
- Trocar esse elemento com o primeiro elemento dessa parte.
- Repetir o procedimento até ordenar o vetor.
- Imprimir o vetor ordenado.
- Resultado esperado: 7 12 19 23 35 48.

Exercício 6

Escreva um programa que leia um número inteiro positivo e verifique se ele é uma potência de 2.

O programa deverá imprimir `1` se o número for uma potência de 2 e `0` caso contrário. Considere como entrada válida um inteiro positivo representável em 32 bits. Exemplo:

Entrada	Saída
8	1
16	1
12	0
1	1
0	0

Exercício 7

Considere a string:

```
.data
texto: .asciiz "Arquitetura de Computadores"
```

Escreva um programa que conte quantas vogais existem na string, desconsiderando diferenças entre letras maiúsculas e minúsculas.

Requisitos:

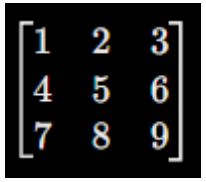
- Percorrer a string caractere por caractere.
- Utilizar lbu para carregar os caracteres.
- Encerrar a leitura ao encontrar \0.
- Contabilizar as vogais a, e, i, o e u, independentemente de maiúsculas ou minúsculas.
- Imprimir a quantidade encontrada.

Exercício 8

Considere a matriz quadrada:

```
.data
matriz: .word 1, 2, 3, 4, 5, 6, 7, 8, 9
```

Considere que ela representa a matriz:



Escreva um programa que calcule e imprima:

- A soma da diagonal principal.
- A soma da diagonal secundária.
- Resultados esperados: diagonal principal = 15; diagonal secundária = 15.

Exercício 9

Considere o vetor:

```
.data
vetor: .byte 10, 20, 10, 30, 10, 40, 20, 50, 10, 60
```

Solicite ao usuário um valor entre 0 e 255 e conte quantas vezes esse valor aparece no vetor.

Requisitos:

- Utilizar lbu para carregar os elementos.
- Comparar cada elemento com o valor informado.
- Contar todas as ocorrências.
- Imprimir o resultado.
- Exemplo: para a entrada 10, a saída será 4.

Exercício 10

Escreva um programa que leia cinco números inteiros e determine o maior deles utilizando um procedimento chamado maximo.

O procedimento deverá receber os endereços ou os valores necessários por meio dos registradores de argumentos, conforme a estratégia escolhida, e retornar o maior valor em $v0.

Requisitos:

- Utilizar jal para chamar o procedimento.
- Utilizar jr $ra para retornar.
- Não calcular o máximo diretamente no programa principal.
- Imprimir o valor retornado.

Atividade Avaliativa

1. Gerar 1 arquivo “.asm” para cada exercício proposto.
2. Compactar os arquivos em “.zip”.
3. Enviar o “.zip” no link correspondente no NEAD.

OBS: A atividade pode ser feita em grupos com até 3 pessoas.